

Temporal Efficacy of Injection-Based and Combined Physical Therapy for Adhesive Capsulitis Across Disease Stages: A Systematic Review and Network Meta-analysis

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Introduction

Adhesive capsulitis, commonly referred to as frozen shoulder (FS), which is characterized by repeated inflammation and fibrosis of the joint capsule, leading to limitation of both passive and active range of motion (ROM) accompanied by pain, thereby affecting shoulder function and quality of life.[1] It occurs in about 2% to 5% of the general population, with a higher occurrence in females, particularly among individuals aged 40–60 years.[2–4] Traditionally viewed as a self-limiting condition, frozen shoulder is increasingly challenged by longitudinal studies showing that a substantial proportion of patients suffer from long-term pain and stiffness.[3]

Primary (idiopathic) adhesive capsulitis develops spontaneously without an obvious precipitating event, while secondary adhesive capsulitis follows trauma, surgery, prolonged immobilization, or occurs in association with systemic conditions such as diabetes or thyroid diseases.[2,3] Prevalence of adhesive capsulitis may exceed 10% in individuals with diabetes and is also less responsive to treatment in these patients.[2–4]

Conservative treatment options for adhesive capsulitis include oral analgesics, home exercises, supervised physical therapy, and injection-based treatment.[2–4] While physical therapy plays a central role in the management of adhesive capsulitis across all disease stages, it alone may be insufficient in the presence of significant pain or marked capsular stiffness, leading clinicians to consider more targeted interventions such as injections.[5,6]

Intra-articular corticosteroid injection (IACI) remains a commonly employed approach in clinical settings.[5] Systematic reviews and meta-analyses have indicated that IACI improves clinical outcomes in the short term. [6–8] However, growing evidence suggests that clinical outcomes are influenced by the anatomical site of injection.[5,7–9] Recent meta-analysis by Liang et al. investigated different corticosteroid injection methods including hydrodilatation (HD), rotator interval (RI), subacromial (SA) or multisite injections, showing that RI injection, HD, and intra-articular injection had equivalent effects on symptom relief.[9]

In addition to corticosteroids, orthobiologics such as platelet-rich plasma (PRP) and hyaluronic acid (HA) have been studied in recent years to treat frozen shoulders. Recent reviews and meta-analyses have shown that these injectants could effectively improve pain and function, and could be considered as alternatives for patients who are not suitable or reluctant to receive IACI.[10,11] Several network meta-analyses have been conducted to identify the best treatment options.[6–8] However, with the emerging number of randomized controlled trials investigating different injection methods and injectants, there remains a need for comprehensive comparison. We aimed to compare the therapeutic effectiveness among various injection treatments using network meta-analysis which enables the indirect comparison of multiple interventions.

Methods

Protocol and registration:

We performed a systematic review and network meta-analysis. The protocol was prospectively registered on PROSPERO (CRD420261323246). We followed the guidelines outlined in the Cochrane Handbook for Systematic Reviews of Interventions and adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Extension Statement for Network Meta-Analyses.[12]

Identification of trials and search method:

We identified eligible studies by searching all fields in MEDLINE (via PubMed), Embase, Cochrane Central Register of Controlled Trials, and related meta-analyses and reviews. Relevant reviews were manually searched for additional references. The search covered records from database inception to February 2026. Text words with key terms of each comparison were entered in all electronic database searches. (search strategies are listed in Supplemental Method S1).

Eligibility criteria:***Participants***

For inclusion, participants had to be older than 18 years with a clinical diagnosis of adhesive capsulitis, frozen shoulder, periarthritis shoulder, painful stiff shoulder. We excluded studies with secondary frozen shoulder (associated with trauma, prior shoulder surgery, stroke, or other identifiable secondary causes) or mixed diagnosis of shoulder pain (eg, painful shoulder syndrome, hemiplegic shoulder pain) were also excluded even if a proportion of them had frozen shoulders. Inclusion of patients with specific conditions (e.g., diabetes) was not an exclusion criterion and was not considered in analyses, provided that the proportion in treatment groups was comparable. However, studies with 100% diabetes cohorts were excluded to minimize clinical heterogeneity.

Interventions

We included any injection-based treatment modality for primary frozen shoulder. Studies comparing different anatomical injection sites (e.g., subacromial [SA] vs. intra-articular [IA] corticosteroid injections) were eligible for inclusion. Furthermore, we excluded studies evaluating multimodal interventions where injection therapy was combined with procedural techniques, such as manipulation under anesthesia (MUA) or arthroscopic capsular release (ACR). Studies enrolling patients with or comparing different types, regimens, dosages, or durations of the same intervention were excluded (eg, different doses of corticosteroid, different volumes or injectate contents for hydrodilatation, or different exercise types).

Outcomes

The primary outcomes were pain and global function. The hierarchy of the scales used to measure each outcome is described in Method S2 in the supplement. The secondary outcome was ROM of external rotation. The post-interventional follow-up time points were defined as short-term (<12 weeks), mid-term (12 weeks to 6 months) and long-term (>6 months).

Types of studies

We included all published randomized controlled trials. Nonrandomized comparative studies, retrospective studies, observational studies, case reports, case series, literature reviews, published conference abstracts, and studies published in languages other than English were excluded.

Data extraction and quality assessment:

Two authors (YHW and AC) performed data extraction. The key characteristics of each eligible article were extracted and inserted in tables in Microsoft Word version 16.43 (Microsoft Corp) to facilitate analysis and presentation. We developed data gathering study information including patient characteristics (number of patients, average age, duration of symptoms), injection modality (injection technique and regimen) and outcome measurement (follow-up period, outcome parameters).

Only treatment that appears 2 or more times will be included in quantitative analysis. A formal risk of bias assessment was performed by two authors (YHW and AC) using the Cochrane Risk of Bias Tool for Randomized Controlled Trials (RoB2).[13] Any disagreements were resolved by consultation with the other reviewer (CHW).

Definitions

The term physiotherapy was used for any supervised, physiotherapist-led, noninvasive treatment (mobilizations, application of ice and heat, diathermy, electrotherapy modalities, standardized exercise program) or when articles explicitly mentioned physiotherapy or physical therapy. These were grouped and analyzed together. Exercises and stretching performed by participants at home (home exercise program) were not included in physiotherapy. Acupuncture, laser therapy, and extracorporeal shock wave therapy (ESWT) were regarded as separate interventions from physiotherapy and were not included.

Interventions with accompanying physiotherapy were grouped and analyzed separately from those without, regardless of intensity and frequency. For example, studies with a treatment group receiving IA corticosteroid plus physiotherapy (e.g., ice packs and diathermy) were included in the intervention category "IA corticosteroid plus physiotherapy"; those with a treatment group receiving only IA corticosteroid (with or without a home exercise program) were included in the "IA corticosteroid" category.

Patients in the following groups were considered control groups and were analyzed together: no treatment, placebo, sham procedures, IA normal saline or lidocaine, simple analgesia, oral corticosteroids, and home exercise alone. Hydrodilatation was defined as injection of over 20 mL of injectate with any mixtures.

Data synthesis and analysis:

Arm-level change scores (post-intervention minus baseline) were computed for each treatment arm. For outcomes where a higher score indicates worse status (e.g., pain), the change score was defined as baseline minus post-intervention, so that all effect estimates were oriented such that positive values reflect clinical improvement. For functional outcomes reported on scales where a higher score indicates better function, the change score was defined as post-intervention minus baseline, with direction verified against each scale's scoring convention. Standard deviations (SDs) of change scores were derived from pre- and post-intervention SDs using the formula:

$$SD_{\text{change}} = \sqrt{SD_{\text{pre}}^2 + SD_{\text{post}}^2 - 2\rho \cdot SD_{\text{pre}} \cdot SD_{\text{post}}}$$

where the pre–post correlation coefficient ρ was assumed to be 0.5, consistent with recommendations from the Cochrane Handbook[14,15]. For the unreported standard deviations (SDs), we calculated the values on the basis of standard errors (SEs), ranges, interquartile ranges, P values, or 95% confidence intervals (CIs) using the method described by the Cochrane Handbook and Wan et al.[14,15] For studies with none of the above statistics reported, we imputed the value based on the weighted mean SDs of the included studies using the same scale, with the sample sizes of the included studies taken into account.

All pain scores originally reported on a 0–100 visual analogue scale (VAS) were rescaled to a 0–10 equivalent by dividing both the mean and SD by 10, to ensure comparability across studies.

A random-effects network meta-analysis was performed within a frequentist framework using the netmeta package (version 3.4) in R (version 4.3). The effect estimates of global function were presented as standardized mean differences (SMDs), those of pain and ROM as mean differences (MDs). The effect estimates were reported with 95% CIs. The reference treatment for all comparisons was the control group (placebo or sham). Treatments appearing in fewer than two independent studies were excluded iteratively. When the resulting network contained disconnected subnetworks, only the largest connected component was retained for analysis.

Follow-up time points were classified into four windows: early short-term (2–6 weeks), late short-term (8–12 weeks), mid-term (13–26 weeks), and long-term (≥ 6 months). All short-term time points were converted to weeks and all mid- and long-term time points to months to ensure consistency. For studies reporting multiple time points within the same window, data from the latest available assessment were used preferentially.

Treatment rankings were estimated using the P-score, the frequentist analogue of the surface under the cumulative ranking curve (SUCRA), where higher values indicate greater probability of being the best treatment.

Assessment of Heterogeneity and Inconsistency

Between-study heterogeneity was quantified using the heterogeneity variance τ^2 , the I^2 statistic, and the total Q statistic, decomposed into within-design (heterogeneity) and between-design (inconsistency) components. The local approach to inconsistency was evaluated using the node-splitting method, which separates direct and indirect evidence for each closed loop and tests whether the two sources yield statistically compatible estimates. A two-sided P value < 0.05 for the direct–indirect comparison was considered indicative of significant inconsistency. For treatment nodes without any direct evidence against the reference, consistency could not be formally tested; these were classified as relying on purely indirect evidence and were noted accordingly in the evidence certainty assessment.

To explore potential sources of heterogeneity, network meta-regression analyses were conducted using the metafor package (version 4.6) (rma.mv, REML estimation) with a two-level random-effects structure (study-level and comparison-level variance components). Three pre-specified moderators were examined: (1) symptom duration (dichotomized as chronic [≥ 6 months] vs. acute/subacute [< 6 months]); (2) injection guidance modality (ultrasound-guided, fluoroscopy-guided, or anatomical landmark-guided, with landmark as the reference category); and (3) baseline pain severity (continuous, 0–10 VAS). Subgroup NMAs stratified by symptom duration and by guidance modality were also performed to characterize treatment effects within each stratum.

Certainty of evidence

The minimum clinically important difference (MCID) for each outcome was estimated as 0.5 for standardized mean difference and 0.5 times the pooled baseline SD across all eligible arms for mean difference, calculated as:

$$SD_{\text{pooled}} = \sqrt{\frac{\sum(n_i - 1) \cdot SD_i^2}{\sum(n_i - 1)}}$$

using a distribution-based approach, applying the 0.5 standard deviation (SD) rule [16].

We defined an intervention with above moderate certainty to have supporting number of studies over 3, confidence interval width < 20%, with head-to-head comparison to control, and no incoherence derived from indirect evidence.

Publication Bias

Small-study effects and potential publication bias were assessed using comparison-adjusted funnel plots, in which each study's centred treatment effect (adjusted for the specific comparison) was plotted against its standard error. Treatments were ordered by their P-score ranking. Funnel plot asymmetry was formally tested using the linear regression method analogous to Egger's test, as implemented in the netmeta package. A two-sided P value < 0.10 was used as the threshold for declaring significant asymmetry, given the typically low power of such tests in NMA settings. Per-comparison funnel plots were additionally generated for all direct comparisons supported by three or more studies.

Result

Figure 1. PRISMA diagram summarizing the search process and included studies

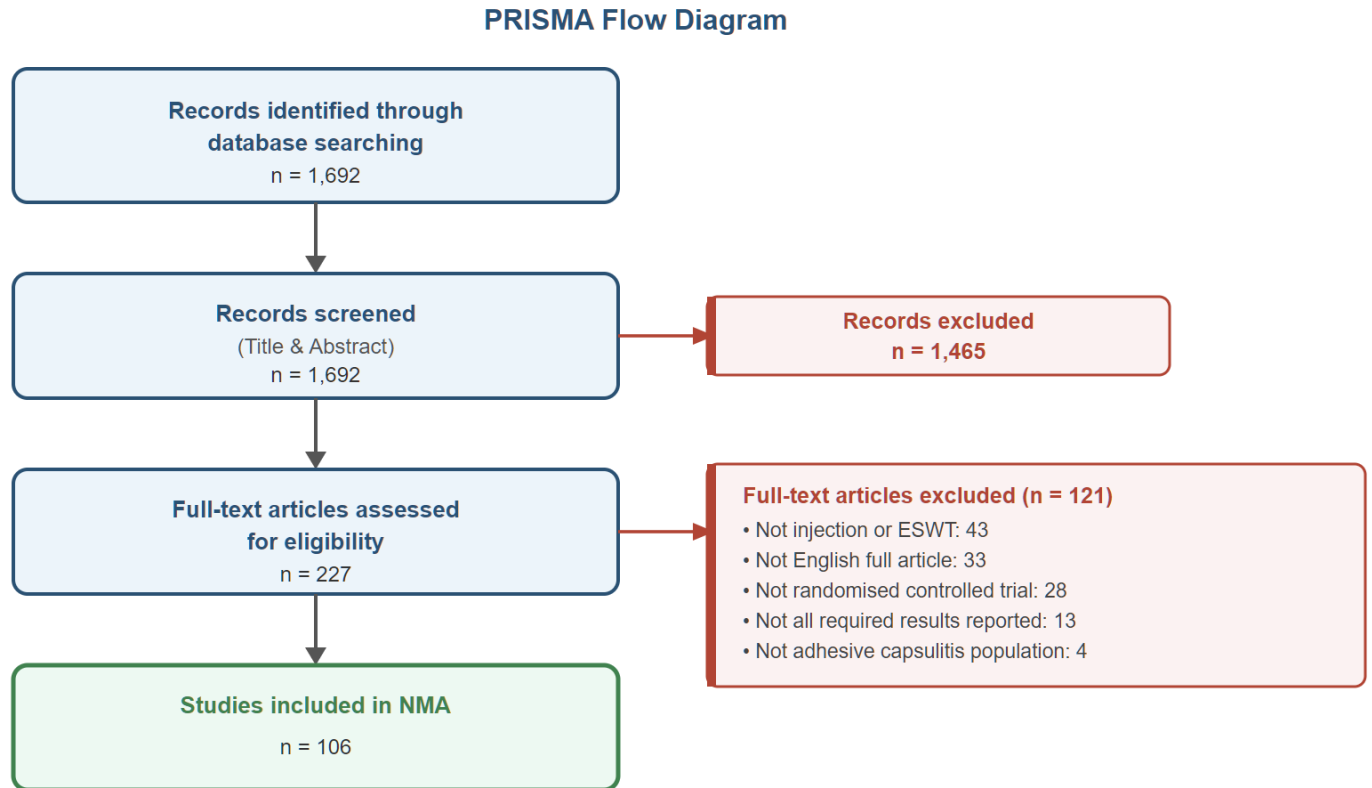


Table 1. Study Characteristics

Source	Injection modality	Physiotherapy, (Y/N)	Participants, n	Mean age, y	Duration of symptoms, m	Follow-up duration	Outcome measures	Included in analysis (Y/N)
Ahn et al. (2015)	IA-CSI, 0.5% lidocaine (4 mL) + triamcinolone 40 mg (1 mL). 2 consecutive therapeutic injections with a 2-week interval if needed	N	57	55.23 ± 4.69	7.23 ± 1.94	1m,3m,6m	SPADI, VNS, passive ROM (ABD, FL, ER)	Y
	IA-HD, 0.5% lidocaine (19 mL) + ketorolac 30 mg (1 mL) + capsular distension. 2 consecutive therapeutic injections with a 2-week interval if needed	N	64	54.63 ± 5.62	7.43 ± 2.32			
Akhtar et al. (2019)	IA-NSAIDs, 60 mg ketorolac + 1% lidocaine with epinephrine (6 mL)	N	80	37.87±1.027	>6	4w	UCLA	N
	IA-HA, 4ml 1% hyaluronic acid + 1% lidocaine with epinephrine (6 mL)	N	80	45.37±5.743	>6			
Anjum et al. (2020)	IA-CSI, 80 mg methylprednisolone + 0.5% bupivacaine to make a 10-mL solution	Y	26	41.12	<6	2w, 6w, 3m	SPADI, VAS, active and passive ROM (ABD, FL, ER, IR)	Y
	None	Y	26	44.46	<6			
Arslan & Çeliker (2001)	IA-CSI, 40 mg methylprednisolone	N	10	55.6 ± 12.2	4.6 ± 1.6	2w,12w	VAS, active and passive ROM	Y

	acetate (1 mL) + 2% lidocaine (1 mL)						(ABD, FL, ER, IR)	
	None	Y	10	56.4 ± 7.1	3.5 ± 1.7			
Bal et al. (2008)	IA-CSI, 40 mg methylprednisolone acetate (1 mL)	N	40	56.9 ± 9.56	1.5–6	2w,12w	SPADI, VAS (night), passive ROM (ABD, FL, ER, IR), UCLA	Y
	Sham injection, 0.9% sodium chloride (1 mL)	N	40	56.3 ± 8.16				
Buchbinder et al. (2004)	IA-HD, 40 mg methylpredni-solone acetate (1 mL) and up to 82 mL normal saline was then injected (total volume 30–90 mL)	N	25	57.2 ± 8.6	118 days (102–194)	3w,6w,12w	SPADI, VAS (night), passive ROM (ABD, FL, ER, IR), UCLA	Y
	Sham injection, standard arthrogram with 6 mL contrast only	N	21	57.5 ± 8.1	114 days (96–402)			
Calis et al. (2006)	IA-HA, sodium hyaluronate 30 mg weekly *2	N	27	59.7 ± 9.8	>1	15d,3m	VAS, Constant score, passive ROM (ABD, ER)	Y
	IA-CSI, 40 mg triamcinolone acetonide	N	26	56.4 ± 11.3	>1			
	None	Y	22	52.3 ± 10.1	>1			
	None	N	20	59.3 ± 6.8	>1			
Carette et al. (2003)	IA-CSI, 40 mg triamcinolone hexacetonide (2 mL)	Y	21	54.9 ± 10.5	22.1 ± 14.9 weeks	6w,3m,6m,12m	SPADI, SPADI Pain subscale, active and passive ROM (ABD, FL, ER), SF-36	Y

	IA-CSI, 40 mg triamcinolone hexacetonide (2 mL)	N	23	55.4 ± 10.0	21.2 ± 11.0 weeks			
	Sham injection, isotonic saline (2 mL)	Y	26	54.2 ± 8.3	20.8 ± 11.2 weeks			
	Sham injection, isotonic saline (2 mL)	N	23	56.5 ± 9.4	20.3 ± 7.3 weeks			
Chan et al. (2024)	IA-CSI, 40 mg triamcinolone + 1% lidocaine (5 mL) + contrast (0.5-1 mL)	Y	21	51 (42-67, total cohort)	>3	3w,6w,12w	SPADI, VAS, passive and active ROM (ABD, FL, ER, IR), CMS, grip strength	Y
	Sham injection, 1% lidocaine (5 mL) + contrast (0.5-1 mL)	Y	21		>3			
Chang et al. (2026)	IA-HA, sodium hyaluronate (3 mL) weekly *3	N	21	59.38 ± 7.5	13.46 ± 17.11	4w,6w,8w,26w	SPADI, passive and active ROM (ABD, FL, ER)	Y
	None	Y	25	59.42 ± 7.43	7.72 ± 8.08			
Chanu et al. (2025)	IA-PRP, PRP (4mL)	N	32	54.75 ± 6.6	3.34 ± 1.1	1w,4w,12w,24w	SPADI, VAS, passive ROM (ABD, FL, ER, IR)	Y
	IA-CSI, 20mg triamcinolone acetonide	N	32	56.03 ± 5.6	3.25 ± 1.1			
Chen et al. (2025)	IA-PRP, PRP (2 mL) + PPP (3 mL) + 1% xylocaine (4 mL)	Y	16	56.44 ± 8.87	4.88 ± 4.35	1m,2m,4m,6m	SPADI, VAS (rest, movement), passive and active ROM (ABD, FL, ER, IR), SDQ, SF-36, self-	Y
	IA-CSI + SA-CSI, 30mg triamcinolone	Y	20	55.70 ± 10.70	6.13 ± 6.88			

	(3mL) + 1% xylocaine (4 mL) + normal saline (2 mL)						assessment	
	Sham injection, 1% xylocaine (4 mL) + normal saline (5 mL)	Y	25	52.93 ± 5.96	5.03 ± 5.83			
Chew et al. (2025)	IA-HD, 40 mg methylprednisolone acetate (4 mL) + 1% lignocaine (5 mL) + normal saline (11 mL)	N	10	65.5 (57.3–77.0)	7.5 (3.5–12.0)	6w	SPADI, VAS, passive and active ROM (ABD, ADD, EX, FL, ER, IR), QuickDASH,	Y
	IA-CSI, 40 mg methylprednisolone acetate (4 mL) + 1% lignocaine (1 mL)	N	10	69.5 (68.0–74.8)	12.0 (6.0–52.5)			
Cho et al. (2016)	IA-CSI, 40 mg triamcinolone acetonide +1% lidocaine (4 mL)	N	42	59.1 ± 7.9	5.3 ± 3.6	3w,6w,12w	ASES, VAS, passive ROM (ABD, FL, ER, IR), SSV	Y
	SA-CSI, 40 mg triamcinolone acetonide +1% lidocaine (4 mL)	N	42	56.0 ± 9.4	4.6 ± 3.5			
	IA-CSI + SA-CSI, 40 mg triamcinolone acetonide +1% lidocaine (4 mL), half in IA, the other half in SA	N	42	54.8 ± 8.3	5.0 ± 4.5			
Cho et al. (2022)	RI-CSI, 40 mg triamcinolone acetonide (1 mL) + 1% lidocaine (3 mL) + water soluble unionized contrast (3 mL) + normal saline (3 mL)	N	45	54.14 ± 8.87	9.35 ± 14.74	3w,6w,12w	ASES, VAS, passive ROM (ABD, FL, ER, IR), SSV	Y

	IA-CSI, 40 mg triamcinolone acetonide (1 mL) + 1% lidocaine (3 mL) + water soluble unionized contrast (3 mL) + normal saline (3 mL)	N	45	55.44 ± 9.93	10.00 ± 11.26			
Choudhary & Gupta (2022)	IA-CSI, 80 mg methylprednisolone mixed + 0.5% bupivacaine to make a 10-mL solution	N	18	46.72	>1	6w,12w,24w	SPADI, VAS, active and passive ROM (ABD, FL, ER, IR)	Y
	IA-HD, 40 mg of corticosteroid + 0.25% local anesthetic (2 mL) + 40–60 ml of normal saline	N	18	45.08	>1			
Dacre et al. (1989)	IA-CSI, 20 mg triamcinolone + 2% lignocaine (1 mL)	Y	22	58.8	>1	6w,26w	VAS, passive ROM (ABD, ER, IR)	Y
	IA-CSI, 20 mg triamcinolone + 2% lignocaine (1 mL)	N	22	55.8	>1			
	None	Y	22	53.0	>1			
Deng et al. (2023)	Multisite-CSI, 40 mg triamcinolone acetonide (1mL) + 2% lidocaine (4mL)	N	30	56.7 ± 12.1	3.3 ± 1.6	4w,8w,12w	ASES, VAS, passive ROM (ABD, FL, ER, IR), CMS	N
	RI-CSI, 40 mg triamcinolone acetonide (1mL) + 2% lidocaine (4mL)	N	29	58.6 ± 11.0	3.7 ± 1.3			

Elnady et al. (2020)	RI-HD, 40 mg methylprednisolone acetate (1 mL) + 2% lidocaine (1 mL) + saline (15 mL)	Y	30	47.6 ± 3.5	8.3 ± 2.68	3m	SPADI, VAS, ROM (ABD, ADD, EX, FL, ER, IR)	N
	IA-HD, 40 mg methylprednisolone acetate (1 mL) + 2% lidocaine (1 mL) + saline (15 mL)	Y	30	45.4 ± 4.9	9.1 ± 2.93			
Elsaman et al. (2025)	SSNB, 0.5% lidocaine (10 mL)	N	22	45.40	32.25	12w	SPADI, VAS, passive ROM (ABD, ER, IR)	Y
	IA-HD, 1% lidocaine (10 mL) + sterile water (20 mL)	N	21	52.05	33.00			
	RI-HD, 1% lidocaine (10 mL) + sterile water (20 mL)	N	19	45.73	29.47			
Foula et al. (2023)	IA-CSI, 40 mg triamcinolone + 0.125% bupivacaine (total 5 mL)	N	15	42.0 (39.0-52.0)	NA	1w,2w,4w,8w	SPADI, VAS (rest, movement), ROM, ICAM-1, hs-CRP	N
	IA-Ozone, oxygen-ozone mixture (15 µg/mL) (10 mL) + 0.125% bupivacaine (5 mL)	N	15	48.0 (39.0-54.0)	NA			
	PRF, 0.125% bupivacaine (5 mL) + intraarticular PRF application using a 10 cm needle - 10 mm active tip (NeuroTherm) for 4 minutes	N	15	46.0 (38.0-57.0)	NA			
Gam et al. (1998)	IA-CSI, 20mg triamcinolone hexacetonide	N	9	47.0 (43-54)	4.5 (3.5-5.8)	3w,6w,12w	VAS (rest, movement), passive ROM (FL, EXT, ABD, ELE, ER),	N

	IA-HD, 20mg triamcinolone hexacetonide + 0.5% Lidocaine (19 mL)	N	13	53.5 (50-63)	5.0 (4.3-6.0)		Analgesic usage	
Gencer-Atalay et al. (2021)	IA-CSI + SSNB, 7mg betamethasone (1 mL) + 0.5% bupivacaine (4 mL) + 0.5% bupivacaine (5 mL, SSNB)	Y	20	55.84 ± 2.19	>3	3w,3m,12m	SPADI, VAS, passive and active ROM (ABD, FL, ER, IR), SF-36	Y
	IA-CSI, 7mg betamethasone (1 mL) + 0.5% bupivacaine (4 mL)	Y	20	51.79 ± 1.58	>3			
Gupta et al. (2022)	IA-PRP, autologous PRP (2 mL)	N	30	47.8 ± 9.56	3.567 ± 1.015	4w,12w,24w	DASH, VAS	Y
	IA-CSI, 40 mg/mL triamcinolone (2 mL).	N	30	46.7 ± 7.13	3.217 ± 0.887			
Haque et al. (2021)	SSNB, 0.5% bupivacaine hydrochloride (10 mL)	N	43	NA	3-12	1w,6w,12w	SPADI, ROM (by modified CMS)	Y
	IA-CSI, 40 mg of triamcinolone (1 mL) and 2% lignocaine hydrochloride (1 mL)	N	43	NA	3-12			
Hsieh et al. (2012)	IA-HA, 20mg hyaluronic acid (2 mL) weekly*3	Y	35	52.6 ± 6.3	5.2 ± 2.6	6w,12w	SPADI, VAS, passive and active ROM (ABD, FL, ER, IR), SDQ, SF-36	N
	None	Y	35	56.4 ± 9.0	3.8 ± 2.6			
Hsieh et al. (2026)	IA-PRP + SA-PRP, PRP (2 mL) + platelet-poor plasma (3 mL) + 1%	Y	30	57 ± 8.2	4.9 ± 4.3	1m,2m,4m,6m	SPADI, VAS, passive and active ROM (ABD, FL, ER, IR), SDQ, SF-	Y

	xylocaine (4 mL) biweekly*2						36, self-assessment	
	IA-CSI + SA-CSI, 30 mg triamcinolone (3 mL) + 1% xylocaine (4 mL) + normal saline (2 mL) biweekly*2	Y	30	55 ± 10.25	6.3 ± 6.4			
	Sham injection, 1% xylocaine (4 mL) + normal saline (5 mL) biweekly*2	Y	30	56.8 ± 7.4	5.4 ± 4.8			
Huang et al. (2024)	IA-HD + SA-CSI, triamcinolone (10 mL, HD) + 1% xylocaine (2 mL, HD) + normal saline (17 mL, HD), triamcinolone (10 mL, SA) + 1% xylocaine (2 mL, SA) biweekly*2	Y + Mobilization	35	53.5 ± 8.1	6.3 ± 4.9	1m,2m,4m,6m	SPADI, SDQ, VAS, passive and active ROM (ABD, FL, ER, IR), SF-36	N
	Sham injection, 1% xylocaine (2 mL) biweekly*2	Y	35	57.5 ± 8.6	5.3 ± 3.6			
Jacob et al. (2024)	IA-HD, 1% lidocaine (3-10 mL, then removed) + rinsed with 0.9% saline 2 to 3 times + cortivazol (Altim) 3.75 mg	Y + Mobilization	33	52.5 ± 11.1	10.5 ± 6.8	15d,6w,3m,6m,12m	SPADI, VAS (rest, movement), passive and active ROM (ABD, FL, ER, IR), DASH	Y
	IA-CSI, 1 steroid injection + cortivazol (Altim) 3.75 mg	Y	33	56.1 ± 8.6	9.2 ± 8.9			
Jadon et al. (2023)	SSNB, 40 mg methylprednisolone acetate + 0.25% bupivacaine (10 mL)	N	48	60.2 ± 10.7	1-12	4w,8w,16w,24w	SPADI, VAS, passive and active ROM (ABD, EXT, FL, ER, IR)	Y

	None	N	48	58.1 ± 10.5	1-12			
Jain et al. (2021)	IA-CSI, 80 mg methylprednisolone (2 mL)	N	50	50.02±8.81	4.4±2.45	1w,4w,12w	SPADI, VAS, passive and active ROM (ABD, FL, EXT, ER, IR)	Y
	SSNB, 40 mg methylprednisolone (1 mL) + 0.5% bupivacaine (10 mL)	N	50	50.52±8.63	4.2±2.03			
Jellad et al. (2020)	IA-CSI, local corticoid (1.5 mL) + sodium and meglumineioxaglate (2 mL) + 1% xylocaine (8 mL)	Y (following injection)	34	55.7 ± 9.8	6 ± 3.5	6w,12w	DASH, VAS (rest, movement), passive and active ROM (ABD, FL, ER)	Y
	IA-CSI, local corticoid (1.5 mL) + sodium and meglumineioxaglate (2 mL) + 1% xylocaine (8 mL)	Y (followed by injection)	46	55.1 ± 7.7	4.4 ± 3.4			
	None	Y	42	55 ± 10.4	5.1 ± 3.3			
Jones et al. (1999)	SSNB, 20 mg triamcinolone acetoneide (0.5 mL) + 0.5% bupivacaine hydrochloride (9.5 mL)	N	15	60 ± 16	NA	1w,3w,7w,12w	Pain (simple and total pain scores), ROM (ABD, ER, IR)	Y
	IA-CSI, 20 mg triamcinolone acetoneide (0.5 mL) + 2% lidocaine hydrochloride (4.5 mL)	N	15	53 ± 10	NA			
Karabaş et al. (2021)	IA-PRP, autologous PRP (3 mL)	N	20	57.3 ± 7.3	6 (3–8)	2w,6w,12w	SPADI, VAS, passive and active ROM (ABD, EXT, FL, ER, IR)	Y
	None	N	20	56.8 ± 5.9	3 (3–7)			

Khallaf et al. (2018)	IA-CSI, 40 mg methylprednisolone acetate (1 mL) + 2% lidocaine (1 mL)	N	20	49.6 ± 8.6	2–12	12w	SPADI, VAS, passive and active ROM (ABD, EXT, FL, ER, IR)	Y
	SA-CSI, 40 mg methylprednisolone acetate (1 mL) + 2% lidocaine (1 mL)	N	20	45.1 ± 8.4	1–12			
Kim et al. (2015)	IA-CSI, 40 mg triamcinolone acetonide (1 mL) + 1% lidocaine (4 mL)	N	46	56.1 ± 7.8	NA	2w,3m	passive ROM (ABD, FL, ER, IR)	Y
	IA-CSI after LC test, 1% lidocaine (2 mL, LC test) then 40 mg triamcinolone acetonide (1 mL) + 1% lidocaine (4 mL)	N	23	58.8 ± 13.0	NA			
	SA-CSI after LC test, 1% lidocaine (2 mL, LC test) then 40 mg triamcinolone acetonide (1 mL) + 1% lidocaine (4 mL)	N	23	52.1 ± 10.9	NA			
Kılıç et al. (2015)	SSNB, triamcinolone (1 mL) + prilocaine (9 mL)	Y	19	55.05 ± 8.29	>1	3w,1m	CSS, Pain (BPI-SF)	Y
	None	Y	22	61.82 ± 9.39	>1			
Konarski et al. (2025)	IA-CSI, contrast 5 mL + 11.4 mg betamethasone (2 mL) + 0.5% marcain (2 mL)	N	38	52 ± 9.8	155 ± 84d	8w	SPADI, VAS (rest and movement), passive ROM (ABD, ER, IR), active ROM (ELE), CSS, ASSES, SST, HBB	Y
	IA-HD, contrast 5 mL + 11.4 mg	N	41	55 ± 8.4	162 ± 121d			

	betamethasone (2 mL) + 0.5% marcain (4 mL) + normal saline until capsular rupture, severe resistance, severe pain or allergic reaction							
Koraman et al. (2021)	Multi-CSI (IA+SA+Biceps+CHL), 40 mg/mL triamcinolone acetonide (2 mL) + 0.5% bupivacaine (4 mL) + saline (34 mL)	Y	38	53.7 ± 7.7	2.7 ± 1.7	1m,3m,6m,12m	ASES, VAS, passive and active ROM (ABD, FL, ER, IR), UCLA	N
	IA-CSI, 40 mg/mL triamcinolone acetonide (1 mL) + 0.5% bupivacaine (2 mL)	Y	38	54 ± 5.6	2.8 ± 1.5			
Kothari et al. (2017)	IA-PRP, PRP (2 mL)	N	65	51.9 ± 10.1	4.1 ± 2.5	3w,6w,12w	QuickDASH, VAS, passive and active ROM (ABD, FL, ER, IR)	Y
	IA-CSI, 80 mg methylprednisolone acetate (2 mL)	N	65	52.7 ± 8.6	5.2 ± 2.8			
	None	Y	65	51.2 ± 11.7	4.7 ± 2.1			
Kraal et al. (2018)	IA-CSI, 40 mg kenacort (1 mL) + lidocaine 1% (4 mL)	Y	10	53.3 ± 3.8	>3	6w,12w,26w	SPADI, VAS (average, night), passive ROM (ABD, FL, ER), RAND-36, satisfaction	Y
	IA-CSI, 40 mg kenacort (1 mL) + lidocaine 1% (4 mL)	N	11	50.4 ± 6.1	>3			
Kulkarni et al. (2021)	SSNB, 0.5% bupivacaine (5 mL)	N	30	49.3 ± 6.1	1–5	4w,8w,12w	QuickDASH, VAS, active ROM (ABD, FL, ER, IR)	Y
	IA-HD, contrast (6 mL) + saline	N	30	50.1 ± 5.1	1–5			

	containing 2% lignocaine (total 30-50 mL)							
Kumar et al. (2024)	IA-CSI + SA-CSI, 40 mg triamcinolone (1 mL, IA) + 2% lignocaine (3 mL, IA); 40 mg triamcinolone (1 mL, SA) + 2% lignocaine (3 mL, SA)	Y	30	52.6 ± 7.8	NA	1m,3m,6m,12m	SPADI, ROM (ABD, ADD, FL, EXT, ER, IR)	Y
	None	Y	30	51.6 ± 8.9	NA			
Lee et al. (2017)	IA-CSI, 40mg triamcinolone acetonide (1 mL) + 1% lidocaine (3 mL)	N	32	NA	>6w (refractory AC)	3w,6w,12w	SPADI, VAS, passive ROM (ABD, FL, EXT, ER, IR)	Y
	IA-HD, 40mg triamcinolone acetonide (1 mL) + 1% lidocaine (6 mL) + normal saline (until capsule dilatation was confirmed)	N	32	NA	>6w (refractory AC)			
Lee et al. (2021)	IA-CSI (low), 20 mg triamcinolone acetonide (0.5 mL) + 2% lidocaine (2 mL) + normal saline (7.5 mL)	N	10	71.30 ± 9.37	4.70 ± 1.64	2w,4w,8w,16w	SDQ, VAS, ROM (ABD, ER), ISF	Y
	IA-CSI (high), 40 mg triamcinolone acetonide (1 mL) + 2% lidocaine (2 mL) + normal saline (7 mL)	N	10	68.56 ± 6.97	5.00 ± 1.94			
	IA-CSI + IA-HA, 20 mg triamcinolone acetonide (0.5 mL) + hyaluronidase	N	10	74.4 ± 4.93	4.30 ± 1.70			

	(1 mL) + 2% lidocaine (2 mL) + normal saline (6.5 mL)							
Lee et al. (2022)	CHL-CSI + IA-CSI (IGHC), 3.5 mg dexamethasone palmitate (1 mL, CHL) + 20 mg triamcinolone acetonide (0.5 mL) biweekly *3 if needed	N	60	53.4 ± 6.3	16.5 ± 7.8w	2m,4m	VAS, passive ROM (ABD, ER, IR)	N
	IA-CSI (PGHR), 3.5 mg dexamethasone palmitate (1 mL, CHL) + 20 mg triamcinolone acetonide (0.5 mL) biweekly *3 if needed	N	60	52.3 ± 6.1	16.6 ± 6.2w			
Lim et al. (2014)	IA-CSI, 40mg methylprednisolone acetate (1 mL) + 1% lidocaine (1 mL)	N	34	53.8 (total average)	7.3 (range 6-13)	2w,12w	CSS, VAS, passive ROM (FL, ER, IR), ASES	Y
	IA-HA, 10 mg/mL sodium hyaluronate (2 mL) weekly *3	N	34					
Lin et al. (2018)	IA-PRP, autologous PRP (2 mL) biweekly *3	N	23	59.8 ± 4.3	NA	1w,1m,3m,6m	UCLA, VAS	Y
	IA-CSI, 25 mg hydrocortisone acetate + 1% procaine biweekly *3	N	23	58.2 ± 4.6	NA			
Lin et al. (2024)	IA-HD, 40 mg triamcinolone acetonide (4 mL) + 2% lidocaine hydrochloride (4 mL) + normal	N	42	54.52 ± 7.92	6.61 ± 5.64	6w,12w	SPADI, VAS (rest, night, movement), active ROM (ABD, FL, ER, IR)	Y

	saline (12 mL)							
	IA-CSI, 40 mg triamcinolone acetonide (4 mL) + 2% lidocaine hydrochloride (4 mL) + normal saline (2 mL)	N	42	52.67 ± 7.20	4.98 ± 3.07			
Lin et al. (2024-2)	SSNB, 20 mg triamcinolone + 1% lidocaine (3 mL) biweekly *2	Y	38	65.8 ± 8.3	5.2 ± 3.1	4w,12w	SPADI, passive and active ROM (ABD, FL, ER, IR), SDQ, SF-36	Y
	IA-CSI, 20 mg triamcinolone + 1% lidocaine (3 mL) biweekly *2	Y	38	66.1 ± 7.2	5 ± 3.4			
Liu et al. (2026)	IA-HD, 40 mg triamcinolone (1 mL) + saline (19 mL) biweekly *3	N	38	55.39 ± 6.75	5.74 ± 1.33	1d,1w,4w,8w,12w	SPADI, VAS, active ROM (ABD, FL, ER, IR)	N
	IA-Dextrose, 5% dextrose (20 mL) biweekly *3	N	37	54.30 ± 4.93	5.64 ± 1.14			
Lorbach et al. (2010)	IA-CSI, 40 mg of triamcinolone + 0.5% bupivacaine (5 mL) + 0.5% mepivacaine (5 mL) *3 in 4-week interval	Y	20	50 ± 8	11	4w,8w,12w,24w,52w	CMS, VAS (pain, function, satisfaction), passive ROM (ABD, FL, ER, IR), SST	Y
	None	Y	20	52 ± 10				
Mansoor et al. (2023)	IA-CSI, 40 mg depomedrol + 0.5% isobaric bupivacaine (6 mL)	Y	20	45.40 ± 6.46	1–6	2w,4w,8w,12w	SPADI (pain, disability)	N
	SSNB, 40 mg depomedrol + 0.5%	Y	20	44.60 ± 7.70	1–6			

	isobaric bupivacaine (6 mL)							
Mardani-Kivi et al. (2022)	None	Y	34	49.20±11.77	>3	2w,4w,6w,12w	SPADI, VAS, ROM (ABD, FL, ER, IR)	Y
	IA-CSI, 80 mg methylprednisolone acetate (2 mL) + 1% lidocaine HCL (8 mL)	Y	32	47.75±10.95	>3			
	SSNB, 80 mg methylprednisolone acetate (2 mL) + 1% lidocaine HCL (8 mL)	Y	31	48.57±10.66	>3			
Maryam et al. (2012)	None	Y	27	53.73±7.49	4.48± 3.37	6w,24w	SPADI, passive and active ROM (ABD, FL, ER)	Y
	IA-CSI + SA-CSI, 60 mg triamcinolone acetonide + lidocaine (3 mL, IA); 20 mg triamcinolone acetonide + lidocaine (1.5 mL, SA)	Y	29	53.71±6.69	6.21± 3.95			
	IA-CSI + SA-CSI, 60 mg triamcinolone acetonide + lidocaine (3 mL, IA); 20 mg triamcinolone acetonide + lidocaine (1.5 mL, SA)	N	31	53.33±7.49	6.83± 3.75			
Medeiros-Filho et al. (2026)	SSNB, 0.25% bupivacaine (6 mL)	Y	145	53.04 ± 8.40	>1	2m,4m,6m	UCLA, VAS, ROM (FL, ER, IR)	Y
	None	Y	145	54.42 ± 9.40	>1			
Mobini et al. (2012)	None	Y	27	53.73±7.49	4.48 ± 3.37	6w,24w	SPADI, passive and active ROM	Y

	IA-CSI + SA-CSI, 60 mg triamcinolone acetonide (IA) + lidocaine (3 mL, IA) ; 20 mg triamcinolone acetonide (SA) + lidocaine (1.5 mL, SA)	Y	29	53.71±6.69	6.21 ± 3.95		(ABD, FL, ER, IR)	
	IA-CSI + SA-CSI, 60 mg triamcinolone acetonide (IA) + lidocaine (3 mL, IA) ; 20 mg triamcinolone acetonide (SA) + lidocaine (1.5 mL, SA)	N	31	53.33±7.49	6.83± 3.75			
Mülkoğlu et al. (2024)	SSNB + HD, 0.5% bupivacaine (6 mL, SSNB) + saline (4 mL, SSNB); 0.5% bupivacaine (4 mL, HD) + saline (12 mL, HD)	N	24	55.8 ± 7.4	5 (2.5 – 6)	1m,3m	QuickDASH, VAS, passive ROM (ABD, FL, ER, IR), HG	Y
	SSNB, 0.5% bupivacaine (6 mL) + saline (4 mL)	N	24	56.2 ± 8.1	5 (4 – 6)			
Mun et al. (2016)	IA-HD, 40 mg triamcinolone (1 mL) + 1% lidocaine (10 mL) + saline solution (30 mL)	Y (+ manipulation)	60	52.1 ± 6.4	6.7 ± 2.3	2w,6w,12w,6m,1y	CSS, VAS, active ROM (FL, ER, IR)	N
	IA-CSI, 40 mg triamcinolone (1 mL) + 1% lidocaine (5 mL)	Y	61	53.9 ± 5.9	6.3 ± 2.1			
Nambi et al. (2025)	IA-CSI, 80mg depomedrol (2 mL) + 1% xylocaine	Y	30	49.18 ± 4.2	6.3 ± 1.6	4w,8w,6m	QuickDASH, VAS, CHL thickness (MRI), TSK-AV, HADS, EQ-5D	Y
	None	Y	30	48.56 ± 4.1	6.5 ± 1.5			

Nasiri et al. (2025)	IA-CSI, 40 mg methylprednisolone acetate (1 mL) + 2% lidocaine (1 mL) + normal saline (3 mL)	N	24	55.67 ± 11.60	7.5 ± 4.39	2w,4w,8w	SPADI, VAS, passive ROM (ABD, FL, ER, IR)	Y
	IA-HD, 2% lidocaine (3 mL) + normal saline (20 mL)	N	24	52.58 ± 6.74	6.33 ± 3.95			
Nazim et al. (2024)	IA-CSI, 80 mg methylprednisolone (2 mL) + 1% lignocaine (2 mL)	Y	36	60.1 ± 6.29	6.7 ± 3.0	1w,3w,6w	SPADI, passive ROM (ABD, ER)	Y
	SSNB, 0.5% bupivacaine (4 mL)	Y	36	58.0 ± 7.83	7.8 ± 3.1			
Ünlü et al. (2020)	IA-PRP, PRP (2 mL) biweekly*3	Y	16	57 ± 8.3	4 (3–9)	1m,3m	SPADI, VAS, ROM, analgesics use	Y
	Sham injection, 0.9% saline (2 mL) biweekly*3	Y	16	56 ± 7.9	6 (3–9)			
Oh et al. (2011)	IA-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (4 mL) + normal saline (4 mL)	N	37	55.7 ± 8.3	6.2 ± 3.6	3w,6w,12w	CSS, VAS, passive ROM (ABD, ER, IR)	Y
	SA-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (4 mL) + normal saline (4 mL)	N	34	58.3 ± 8.9	6.9 ± 3.4			
Oh et al. (2021)	Sham injection, saline (4 mL) + contrast media (4 mL)	N	15	49.4 ± 4.9	7.4 ± 5.9	1d,1w,1m,3m,6m	SPADI, VAS (rest, movement, night, average, worst), passive and active ROM (ABD, FL, ER, IR), ASES, CSS, UCLA, DASH, muscle	Y
	IA-CSI, 40 mg triamcinolone acetate (1 mL) + saline (3 mL) +	N	15	52.3 ± 8.5	7.6 ± 8.6			

	contrast media (4 mL)						strength (SST, IST, SB)	
	IA-HA, hyaluronic acid (2 mL) + saline (2 mL) + contrast media (4 mL)	N	15	54.5 ± 5.1	6.8 ± 8.3			
	IA-CSI + IA-HA, 40 mg triamcinolone acetonide (1 mL) + hyaluronic acid (2 mL) + saline (1 mL) + contrast media (4 mL)	N	15	53.5 ± 7.5	10.2 ± 8.9			
Ozbayrak et al. (2025)	IA-HD + CHL release, 40 mg triamcinolone acetonide (1 mL) + 2% lidocaine hydrochloride (4 mL) + 0.9% saline (up to 50 mL); normal saline (2 mL, CHL)	N	30	56.9 ± 7.8	10.9 ± 14.1	1w,1m,3m	SPADI, VAS (rest, movement, night), ROM (ABD, FL, ER, IR)	N
	IA-HD, 40 mg triamcinolone acetonide (1 mL) + 2% lidocaine hydrochloride (4 mL) + 0.9% saline (up to 50 mL)	N	33	55.8 ± 10.4	8.2 ± 12.7			
Parashar et al. (2020)	None	Y	20	57.90 ± 6.14	NA	12w	SPADI, ROM (ABD, ER, IR, TOTAL)	Y
	SSNB, 80 mg methylprednisolone acetate (2 mL) + 0.5% bupivacaine (10 mL)	Y	20	57.20 ± 6.40	NA			
	IA-CSI, 80 mg methylprednisolone	Y	20	60.35 ± 5.54	NA			

	acetate (2 mL) + 0.5% bupivacaine (3 mL) + normal saline (7 mL)							
Park et al. (2000)	IA-HD, 40 mg triamcinolone acetanide (1 mL) + 2% lidocaine (8 mL) + saline (4 mL) + contrast media (12 mL)	N	28	54.9 ± 8.46	5.49 ± 4.08	1w,1m	VAS, active ROM (ABD, FL, ER, IR)	Y
	IA-CSI, 40 mg triamcinolone acetanide (1 mL) + 2% lidocaine (2 mL) + contrast media (4 mL)	N	27	57.7 ± 9.28	4.67 ± 2.79			
Park et al. (2013)	IA-CSI, 40 mg triamcinolone (1 mL) + 0.5% lidocaine (4 mL) biweekly*3	N	45	55.23 ± 4.69	5.31 ± 1.64	2w,6w	SPADI, VAS, passive ROM (ABD, FL, ER)	Y
	IA-HD, 20 mg high molecular weight sodium hyaluronate (2 mL) + 0.5% lidocaine (18 mL) biweekly*3	N	45	56.33 ± 5.92	5.33 ± 3.62			
Park et al. (2014)	HD-CSI, 40 mg of triamcinolone + 1% lidocaine (3 mL) + normal saline (10 ml)	Y + Mobilization	16	56 ± 7.6 (total cohort)	3–9	4w	SPADI, VAS, active ROM (ABD, FL, ER, IR), CMS	Y
	None	Y + Mobilization	14		3–9			
	HD-CSI, 40 mg of triamcinolone + 1% lidocaine (3 mL) + normal saline (10 ml)	Y	12		3–9			

	None	Y	11		3–9			
Paruthikunnan et al. (2020)	IA-HD, 80 mg methylprednisolone acetate (2 mL) + 0.25% bupivacaine (12-18 mL)	Y	44	56.5 (range: 40–77)	5.0 (range: 1–12)	1.5m,3m,6m	SPADI, VAS	Y
	IA-CSI, 80 mg methylprednisolone acetate (2 mL) + 0.25% bupivacaine (2 mL)	Y	44	54.9 (range: 39–68)	5.1 (range: 2–12)			
Prasad et al. (2024)	Sham injection, saline (4 mL) + contrast media (4 mL)	N	10	48.2 ± 4.4	6.3 ± 5.1	1d,1w,1m,3m,6m	SPADI, VAS (rest, movement, night, average, worst), passive and active ROM (ABD, FL, ER, IR), muscle strength (SST, IST, SB)	Y
	IA-CSI, 40 mg triamcinolone acetonide (1 mL) + saline (3 mL) + contrast media (4 mL)	N	10	54.6 ± 6.5	6.7 ± 6.8			
	IA-HA, hyaluronic acid (2 mL) + saline (2 mL) + contrast media (4 mL)	N	10	53.1 ± 8.2	8.6 ± 5.8			
	IA-CSI + IA-HA, 40 mg triamcinolone acetonide (1 mL), hyaluronic acid (2 mL) + saline (1 mL) + contrast media (4 mL)	N	10	50.3 ± 10.5	9.9 ± 8.3			
Prestgaard et al. (2015)	IA-CSI, 3.5 mg lidocaine (3.5 mL, RI); 20 mg triamcinolone hexacetonide (1 mL, IA) + 2.5 mg lidocaine (2.5 mL, IA)	N	42	53.2 ± 6.9	15.1 ± 4.6w	3w,6w,12w,26w	SPADI, VAS (rest, night), active ROM (ABD, FL, ER), EQ-5D	Y

	IA-CSI + RI-CSI, 10 mg triamcinolone hexacetonide (0.5 mL, IA) + 3 mg lidocaine (3 mL, IA); 10 mg triamcinolone hexacetonide (0.5 mL, RI) + 3 mg lidocaine (3 mL, RI)	N	40	55.0 ± 7.2	15.0 ± 5.9w			
	Sham injection, 3.5 mg lidocaine (3.5 mL, IA); 3.5 mg lidocaine (3.5 mL, RI)	N	40	55.4 ± 6.5	15.0 ± 5.6w			
Pushpasekaran et al. (2017)	IA-CSI, 40 mg methylprednisolone acetate + 2% lignocaine (2 mL)	Y	43	56.4 ± 4.32	15.2 ± 13.74	3w,6w,6m	CONSTANT score (pain, ADL, ROM, strength)	Y
	IA-CSI + SA-CSI, 40 mg methylprednisolone acetate + 2% lignocaine (2 mL) + normal saline (8 mL)	Y	42	56.24 ± 5.42	14.82 ± 13.65			
Ranalletta et al. (2016)	IA-CSI, 12 mg + 1% lidocaine (2 mL)	Y	38	62.9 ± 12.2	12 (8–16)w	2w,4w,,8w,12w	ASES, VAS, passive ROM (ABD, FL, EXT, ER, IR), CMS, QuickDASH	Y
	None	Y	36	63.9 ± 6.91	12 (8–24)w			
Ravishankar et al. (2023)	IA-PRP, autologous PRP (2 mL)	N	28	56.04 ± 6.43	6.0 ± 1.33	6w,12w,18w	SPADI, VAS, passive ROM (ABD, FL, EXT, ER)	Y
	IA-CSI, 40 mg triamcinolone + 2% lidocaine (1 mL)	N	28	55.75 ± 7.29	6.8 ± 1.12			
Ryans et al. (2005)	IA-CSI, 20 mg triamcinolone (1 mL) + normal saline (2 mL)	Y	20	56.3 ± 6.4	14.2 ± 4.4w	6w,16w,24w	SDQ, VAS(rest, global), passive ROM (ER), SF-36	Y

	IA-CSI, 20 mg triamcinolone (1 mL) + normal saline (2 mL)	N	19	52.3 ± 9.3	12.2 ± 5.3w			
	Sham injection, normal saline (3 mL)	Y	20	52.6 ± 7.7	14.4 ± 4.4w			
	Sham injection, normal saline (3 mL)	N	19	55.2 ± 9.4	14.9 ± 3.7w			
Reza et al. (2013)	IA-CSI, 40 mg triamcinolone acetonide (1 mL)	N	50	58 ± 11	119 ± 11d	2d,12w	VAS, ROM (ABD, FL, EXT, ER, IR)	Y
	IA-HD, 40 mg triamcinolone acetonide (1 mL) + 2% xylocaine (2 mL) + contrast (20 mL) + normal saline (27 mL)	N	50	61 ± 9	112 ± 14d			
Sahillioğlu et al. (2025)	IA-ozone, 15 µg/ mL (O2-O3) (20 mL) twice/weekly *4	N	20	54.85 ± 6.35 (total cohort)	>3	4w,12w	SPADI, VAS (rest, movement, night), passive and active ROM (ABD, ADD, FL, EXT, ER, IR)	N
	IA-CSI, 7 mg/1 mL betamethasone	N	20		>3			
Saleem et al. (2026)	IA-CSI, 80 mg methylprednisolone (2 mL) + 1% lignocaine (2 mL)	Y	30	51.9 ± 8.2	NA	1w,6w,12w	SPADI, passive ROM (ABD, FL, ER)	Y
	SSNB, 0.5% bupivacaine (4 mL)	Y	30	52.7 ± 8.7	NA			
Sam et al. (2023)	Multi-dextrose, not specified	N	19	58.16 ± 6.344	NA	6w,12w	WOMAC, VAS, passive ROM (ABD, ADD, FL, EXT, ER, IR), MMP-1/TIMP-1 ratio	N
	Sham injection, not specified	N	20	57.6 ± 10.704	NA			
Sarı et al. (2024)	SSNB, 40 mg of triamcinolone acetonide (1 mL) + 0.5%	N	30	60.0 ± 10.2	6.4 ± 5.9	1w,4w,12w	SPADI, VAS, passive and active ROM (ABD, FL, EXT, ER, IR)	Y

	bupivacaine hydrochloride (9 mL)							
	IA-CSI, 40 mg of triamcinolone acetate (1 mL) + 0.5% bupivacaine hydrochloride (9 mL)	N	30	57.6 ± 9.5	8.2 ± 6.8			
Sayed et al. (2022)	IA-CSI, 80 mg triamcinolone (2 mL) + 2% lidocaine (4 mL)	N	20	51.50 (IQR: 42.0–60.0)	5.0 (IQR: 3.0–7.0)	1w,3w,6w	SPADI, VAS, passive and active ROM (ABD, FL, ER)	Y
	IA-HA, hyaluronic acid (2 mL) + 2% lidocaine (4 mL)	N	20	51.00 (IQR: 46.0–55.0)	5.50 (IQR: 5.0–8.0)			
	IA-CSI + SSNB, 80 mg triamcinolone (2 mL, IA) + 2% lidocaine (4 mL, IA); 0.5% bupivacaine (10 mL, SSNB)	N	20	55.00 (IQR: 50.0–64.0)	5.0 (IQR: 3.0–6.0)			
	IA-HA + SSNB, hyaluronic acid (2 mL, IA) + 2% lidocaine (4 mL, IA) ; 0.5% bupivacaine (10 mL, SSNB)	N	20	55.00 (IQR: 50.0–59.0)	6.00 (IQR: 4.0–7.0)			
Schiltz et al. (2022)	SSNB, 10 mg ropivacaine (5mL) *3	N	19	54 ± 6.4	13 ± 4.0w	1w,2w,4w	CSS, VAS, ROM (ABD, FL, ER, IR)	Y
	Sham injection, normal saline (5 mL) *3	N	16	55 ± 11.9	15 ± 4.9w			
Schydrowsky et al. (2012)	IA-CSI, 40 mg methylprednisolone acetate + 1% lidocaine (4 mL) biweekly*3	N	8	51 (37–64)	>3	2w,4w,8w,12w,24w	SPADI, passive and active ROM (ABD, FL, ER), CSS, SDQ	N
	SC-TNF, adalimumab (1 mL)	N	10	51 (41–67)	>3			

	biweekly*3							
Sensoz et al. (2024)	IA-CSI, 20 mg triamcinolone acetonide (1 mL) and 2% prilocaine (1 mL)	N	26	54.14 ± 8.4	5.13 ± 3.6	1w,3w,6w	ASES, VAS, active ROM (ABD, FL, ER, IR), QuickDASH	N
	CHL-CSI, 20 mg triamcinolone acetonide (1 mL) and 2% prilocaine (1 mL)	N	28	52.54 ± 6.7	4.19 ± 2.5			
Shahzad et al. (2021)	IA-PRP, autologous PRP (2 mL)	N	102	52.41 ± 2.67	5.57 ± 1.49	12w	UCLA, VAS, ROM (ABD, FL, ER, IR)	Y
	IA-CSI, 80 mg methylprednisolone acetate (2 mL)	N	100	53.0 ± 3.74	4.86 ± 1.50			
Shanahan et al. (2022)	IA-CSI + SSNB, 20mg triamcinolone (IA) + 2% local anaesthetic (2 mL, IA); 40 mg methylprednisolone (SSNB) + 10 mg 0.5% bupivacaine (SSNB)	Y	27	51.9 ± 8.6	8.7 ± 7.2	3m,6m,9m	PADI, VAS, passive and active ROM (ABD, FL, ER), time to resolution	Y
	IA-CSI + Sham injection, 20 mg triamcinolone + 2% local anaesthetic (2 mL); normal saline (2 mL, SC)	Y	27	55.7 ± 8.3	10.6 ± 8.1			
Sharma et al. (2016)	IA-CSI, 20 mg triamcinolone (1 mL) + 30 mg lidocaine (3 mL)	N	36	52 ± 8.3	7.5 (range: 2.0–18.0)	4w,8w,12m	SPADI, VAS, passive ROM (ABD, ER, IR)	Y
	IA-HD, 20 mg triamcinolone (1 mL) +	N	34	53 ± 9.2	7.0 (range: 3.0–			

	30 mg lidocaine (3 mL) + normal saline (total 8-20 mL)				37.0)			
	None	N	36	54 ± 6.9	6.0 (range: 3.0–24.0)			
Singh et al. (2024)	CHL-CSI, 80 mg methylprednisolone acetate (2 mL) + 2% lignocaine (2 mL)	Y	20	54 ± 8	5.3 ± 1.7	6w,12w	SPADI, VAS, ROM (ABD, FL, ER, IR) , CMS	N
	Sham injection, 0.25% bupivacaine (2 mL) + 2% lignocaine (2 mL)	Y	19	51 ± 8	5.5 ± 2.3			
Singha Roy et al. (2025)	SSNB, 40 mg triamcinolone (1 mL) + 0.5% bupivacaine (5 mL)	N	28	NA	>1	1w,6w	SPADI, VAS, ROM (ABD, FL, ER, IR)	Y
	SSNB + IA-HD, 40 mg triamcinolone (1 mL, SSNB) + 0.5% bupivacaine (5 mL, SSNB); 0.5% bupivacaine (5 mL, IA-HD) + 0.9% normal saline (15-25 mL, IA-HD)	N	28	NA	>1			
	None	N	28	NA	>1			
Somisetty et al. (2023)	IA-PRP, PRP (4 mL)	N	34	58.3 ± 8.1	NA	2w,4w,8w,12w,24w	SPADI, VAS, QuickDASH	Y
	IA-CSI, 80 mg methylprednisolone acetate (2 mL) + normal saline (2 mL)	N	34	58.5 ± 7.7	NA			

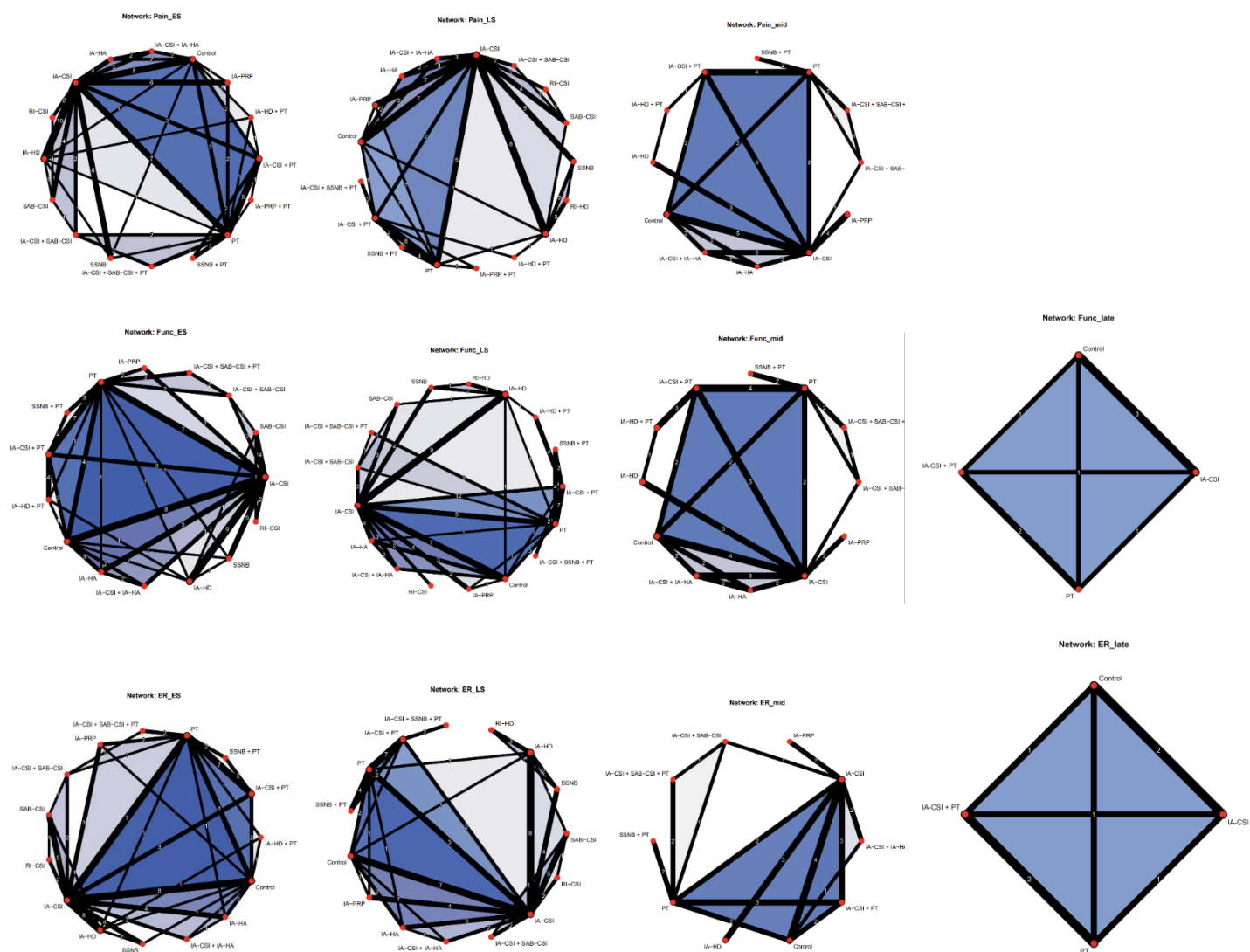
Sun et al. (2017)	RI-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (2 mL)	N	33	52.6 ± 4.4	14.4 ± 4.3w	4w,8w,12w	CSS, VAS, passive ROM (ABD, FL, ER, IR), DASH	Y
	IA-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (2 mL)	N	30	55.1 ± 3.4	15.2 ± 5w			
	SA-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (2 mL)	N	34	54.2 ± 5.1	15.1 ± 4.8w			
Swaroop et al. (2023)	IA-HD, 80 mg triamcinolone (2 mL) + 2% lignocaine (2 mL) + 0.2% ropivacaine (2 mL) + normal saline (until capsule distension)	Y	54	57.57 ± 8.53	3.56 ± 1.20	2w,6w,3m	SPADI, days of analgesia	Y
	IA-CSI, 80 mg triamcinolone (2 mL) + 2% lignocaine (2 mL) + 0.2% ropivacaine (2 mL)	Y	54	58.13 ± 6.39	3.50 ± 1.14			
Thamyongkit et al. (2026)	IA-HA + IA-NSAIDs, 20 mg hyaluronic acid (Ostenil) (2 mL) + 30 mg Ketolorac (1 mL) + 20mg 1% lidocaine (2 mL)	N	28	56.4 ± 7.8	16w (IQR: 8,24)	2w,6w,12w,24w	Thai SPADI, VAS, passive ROM (ABD, ADD, FL, EXT, ER, IR), analgesics use	N
	IA-NSAIDs, 30 mg Ketolorac (1 mL) + 20mg 1% lidocaine (2 mL) + normal saline (2 mL)	N	28	57.4 ± 9.7	12w (IQR: 8,24)			
Thu et al. (2020)	IA-PRP, PRP (4 mL) + 2% lidocaine (1 mL)	N	32	52.84 ± 6.92	<3	1w,3w,6w	DASH, VAS, ROM (ABD, FL, ER)	Y

	None	Y	32	57.17 ± 6.93	<3			
Tveitå et al. (2008)	IA-HD, 20 mg triamcinolone acetanide (2 mL) + 4 mg/mL bupivacaine (4 mL) + contrast (4 mL) + normal saline (10 mL) biweekly*3	N	39	52 ± 7	7 ± 4	6w	SPADI, passive and active ROM (ABD, FL, ER, IR)	Y
	IA-CSI, 20 mg triamcinolone acetanide (2 mL) + 5 mg/mL bupivacaine (3-4 mL) + contrast (3-4 mL) biweekly*3	N	37	51 ± 6	7 ± 4			
van der Windt et al. (1998)	IA-CSI, 40 mg triamcinolone acetanide, no more than 3 in 6 weeks	N	53	57.3 ± 10.2	NA	3w,7w,13w,26w,52w	SDQ, VAS(day, night), passive ROM (ABD, ER), satisfaction	N
	None	Y	56	60.2 ± 10.7	NA			
Verma et al. (2019)	SSNB, 0.5% bupivacaine (4 mL)	N	29	51 (43-60)	3 (4-6)	1w,3w,6w	SPADI, passive ROM (ABD, FL, ER)	Y
	IA-CSI, 80 mg triamcinolone acetanide (2 mL)	N	31	55 (49-60)	3 (4-6)			
Wang et al. (2021)	RI-HD, 40 mg triamcinolone acetanide (4 mL) + 2% lidocaine hydrochloride (4 mL) + normal saline (12 mL)	N	32	52.40 ± 6.37	6.12 ± 3.15	6w,12w	SPADI, VAS (rest, night, movement), ROM (ABD, FL, ER, IR)	Y
	IA-HD, 40 mg triamcinolone acetanide (4 mL) + 2% lidocaine	N	32	53.96 ± 7.02	6.65 ± 4.10			

	hydrochloride (4 mL) + normal saline (12 mL)							
Wu et al. (2024)	IA-HD, hyaluronic acid (2 mL) + 2% lidocaine hydrochloride (4 mL) + normal saline (14 mL) biweekly*3	Y	31	52.94 ± 7.96	9.19 ± 10.15:	6w,12w	SPADI, VAS (rest, night, movement), ROM (ABD, FL, ER, IR), CMS, SF-36	Y
	None	Y	31	54.35 ± 9.77	10.58 ± 10.32			
Yadav et al. (2021)	IA-PRP, autologous PRP (volume not mentioned)	N	30	NA	NA	1w,4w,12w	CMS, VAS	Y
	IA-CSI, 80 mg methylprednisolone (2 mL)	N	30	NA	NA			
Yoon et al. (2013)	IA-CSI, 40 mg triamcinolone acetone (4 mL) + 1% lidocaine (1 mL)	N	20	54.2 ± 5.1	5.5 ± 2.5	1w,3w,6w,12w	SPADI, VAS, passive ROM (ABD, FL, EXT, ER, IR)	Y
	IA-CSI, 20 mg triamcinolone acetone (2 mL) + 1% lidocaine (3 mL)	N	20	52.2 ± 3.8	4.7 ± 2.1			
	Sham injection, 1% lidocaine (5 mL)	N	13	55.9 ± 3.1	5.1 ± 3.1			
Yoon et al. (2016)	IA-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (4 mL) + normal saline (5 mL)	N	30	53 ± 8	9 ± 6	1m,3m,6m	CSS, VAS, passive ROM (FL, ER, IR), SST	Y
	SA-CSI, 40 mg triamcinolone (1 mL)	N	30	57 ± 7	9 ± 5			

	+ 2% lidocaine (4 mL) + normal saline (5 mL)							
	IA-HD, 40 mg triamcinolone (1 mL) + 2% lidocaine (4 mL) + normal saline (40 mL)	N	30	54 ± 9	9 ± 6			
Zare et al. (2025)	None	Y	24	57.79 ± 9.52	6.37 ± 4.21	6w	SPADI, VAS, passive ROM (ABD, ER, IR, Scaption)	Y
	IA-CSI, 40 mg triamcinolone (1 mL) + 2% lidocaine (9 mL)	N	24	56.21 ± 6.34	5.87 ± 3.76			
Zhang et al. (2022)	IA-HD, 2% lidocaine (5 mL) + normal saline (30 mL)	Y	90	54.97 ± 4.8	5.26 ± 0.96	1w,2w,12w,24w	SPADI, VAS, active ROM (ABD, FL, EXT, ER, IR, ELE), CMS	Y
	IA-HD, 2% lidocaine (5 mL) + normal saline (30 mL)	N	90	52.21 ± 7.77	6.46 ± 0.57			

Figure 2. Network diagram of eligible interventions

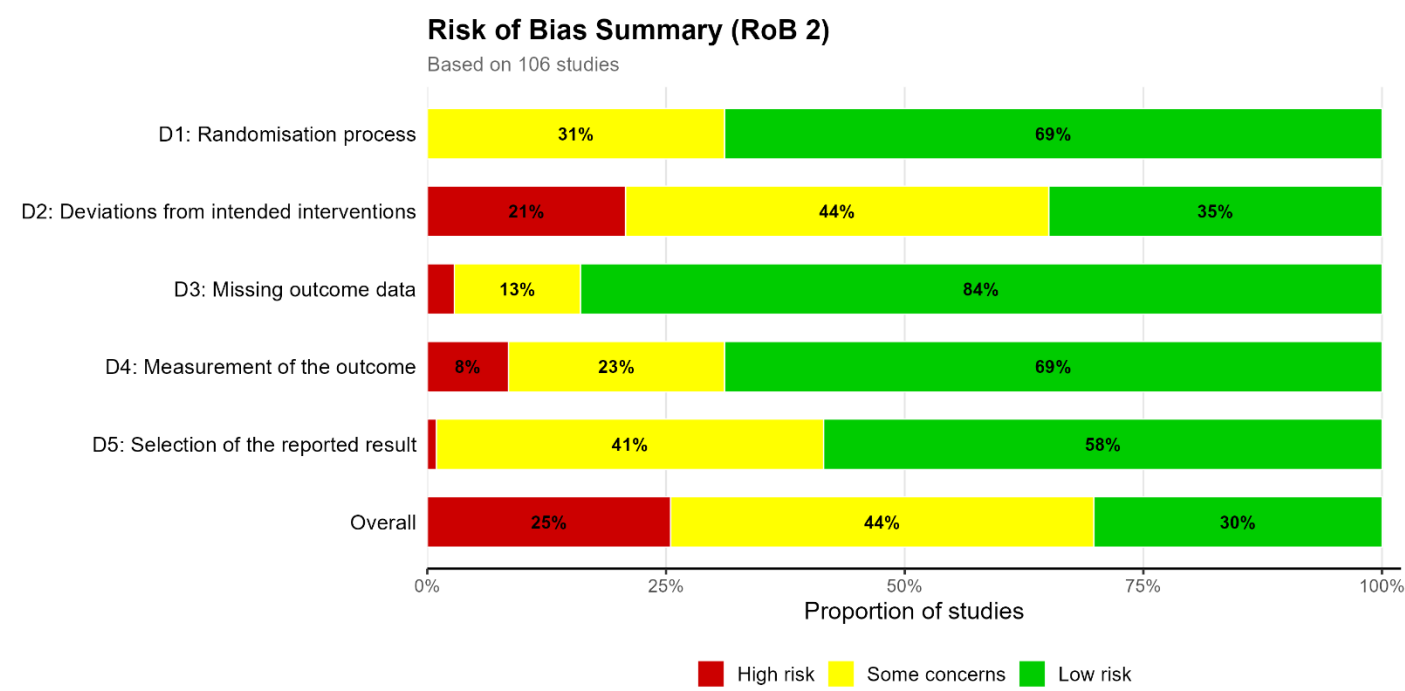


Each node represents a treatment intervention, with node size proportional to the number of participants. Each edge (line) connecting two nodes represents a direct comparison between treatments, with line thickness proportional to the number of studies contributing to that comparison.

Characteristics and risk of bias of the included studies

After the search, we included 106 studies in our qualitative analysis. A total of 86 randomized controlled trials (RCTs) comprising 5611 patients met the formal inclusion criteria for quantitative network meta-analysis. The inclusion process was summarized in **Figure 1**. The final network architecture evaluated 9 foundational monotherapies: corticosteroid injection (CSI), hydrodilatation (HD), platelet-rich plasma (PRP), hyaluronic acid (HA) via intra-articular (IA) or rotator interval (RI) approaches, subacromial bursa (SAB) injection, suprascapular nerve block (SSNB), and physical therapy (PT). All of the characteristics in our included studies were organized as in **Table 1**. Multi-modal regimens were treated as distinct intervention nodes within the global model (**Figure 2**).

Figure 3. Risk of bias summary across the included randomized controlled trials.

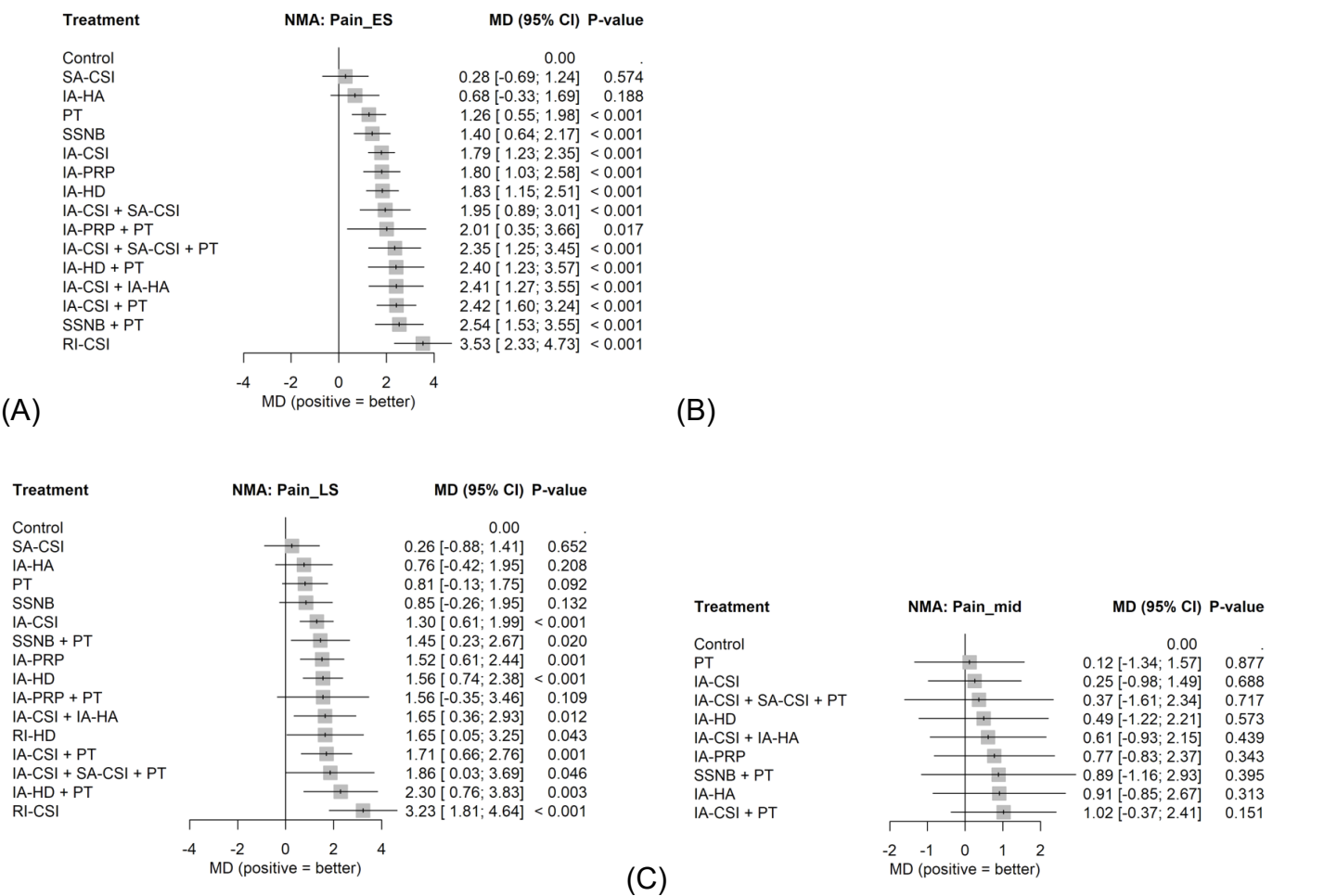


Risk of bias was assessed using the Cochrane Risk of Bias 2 (RoB 2) tool across five domains: D1, randomization process; D2, deviations from intended interventions; D3, missing outcome data; D4, measurement of the outcome; D5, selection of the reported result. Bars represent the proportion of studies rated as low risk (green), some concerns (yellow), or high risk (red). The overall risk of bias profile was predominantly moderate, with D2 and D5 identified as the most problematic domains, largely attributable to the open-label nature of injection-based interventions and selective outcome reporting.

Figure 3 summarizes the risk of bias across 106 RCTs using the RoB 2 tool. The overall profile was moderate, with only 30% of studies rated low risk and 25% high risk, while the majority (44%) raised some concerns. The most problematic domain was deviations from intended interventions (D2), where 21% of studies were rated high risk, which was an expected limitation given the inherent difficulty of blinding participants in injection-based trials. Selection of the reported result (D5) was also concerning, with 41% rated as some concerns, suggesting potential outcome reporting bias. Conversely, missing outcome data (D3) and randomization (D1) were relatively well-handled. These findings indicate some risk of bias across the evidence base, which informed further. Specific study-level breakdowns and domain-specific assessments were detailed in **Supplemental Table S1**.

Pain Relief

Figure 4. Forest plot of network meta-analysis estimates for pain relief



Forest plot displaying the random-effects network meta-analysis estimates for all eligible interventions compared to the reference group (Control). Effect sizes are presented as mean differences (MD) with 95% confidence intervals. A positive value indicates a beneficial effect (improvement) relative to Control.

Early Short-term (2–6 Weeks)

The absolute statistically highest-ranked intervention for early short-term pain relief was rotator interval corticosteroid injection (RI-CSI) (vs Control: MD=3.49, 95% CI, 2.24 to 4.74 in pain relief), which significantly outperformed multi-modal alternatives such as hydrodilatation combined with physical therapy (HD+PT). In contrast, among the interventions supported by moderate-to-high certainty evidence, IA-CSI + PT (vs Control: MD=2.42, 95% CI 1.60 to 3.24) represented the most clinically optimal and stable regimen (**Figure 4**).

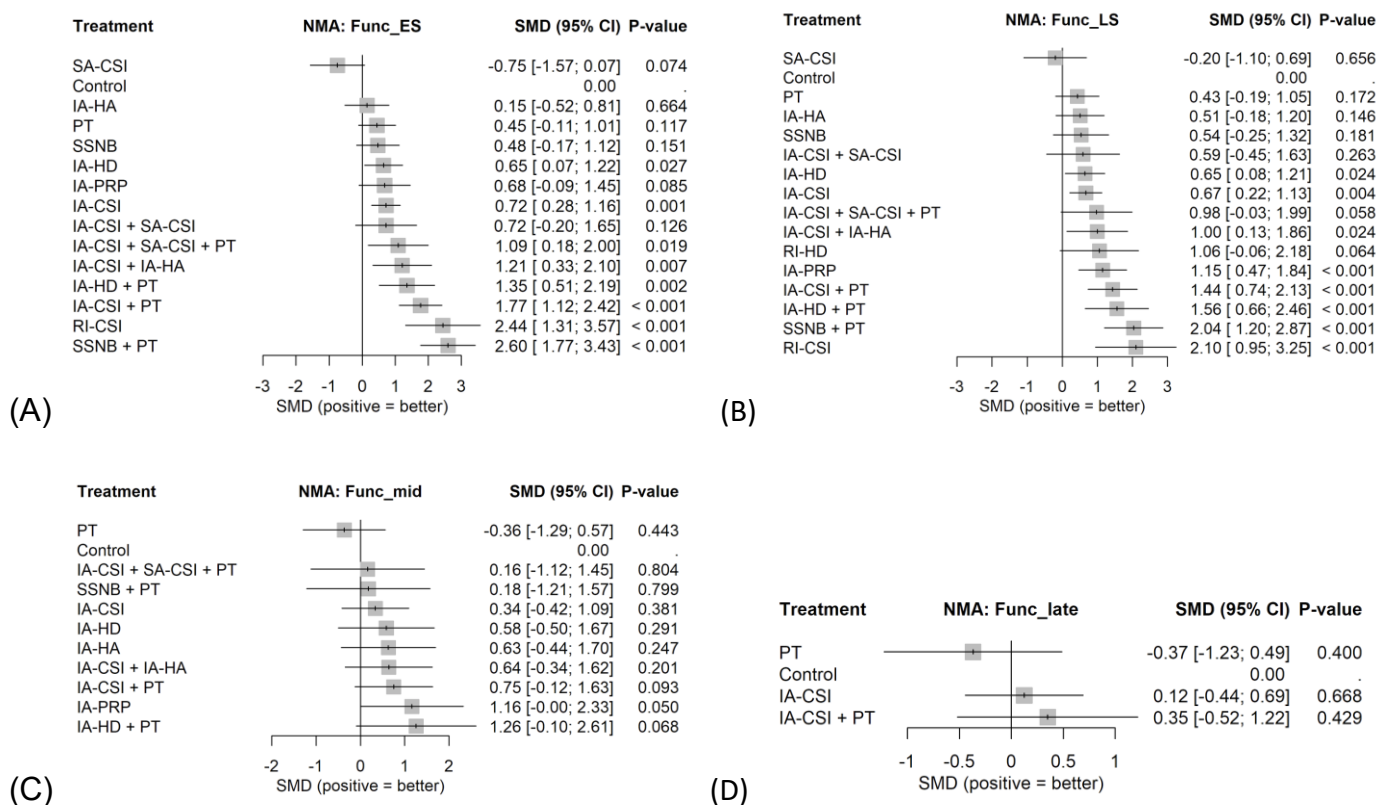
Late Short-term (8–12 Weeks)

RI-CSI maintained its absolute statistical supremacy into the late short-term window, demonstrating the highest superficial ranking probability (vs Control: MD=3.05, 95% CI 1.77 to 4.33). However, when evaluating the network under the constraint of moderate-to-high certainty evidence, no clinically stable intervention was found.

Mid-term (13–26 Weeks)

The therapeutic efficacy profile shifted toward multi-modal intervention at the mid-term horizon, where IA-CSI+PT achieved the highest trend of efficacy hierarchy.

Figure 5. Forest plot of network meta-analysis estimates for functional recovery



Forest plot displaying the random-effects network meta-analysis estimates for all eligible interventions compared to the reference group (Control). Effect sizes are presented as mean differences (MD) or standardized mean differences (SMD) with 95% confidence intervals. A positive value indicates a beneficial effect (improvement) relative to Control. Treatments are sorted by effect size.

Functional Recovery

Early Short-term (2–6 Weeks)

During the early short-term window, suprascapular nerve block combined with physical therapy (SSNB+PT) (vs Control: SMD=2.60, 95% CI 1.77 to 3.43) and corticosteroid injection to the rotator interval (RI-CSI) (vs Control: SMD=2.44, 95% CI 1.31 to 3.57) secured the absolute highest statistical rank. When restricted to moderate-to-high certainty evidence, IA-CSI+PT (vs Control: SMD=1.77, 95% CI 1.12 to 2.42) was the most optimal regimen. The forest plots were demonstrated in **Figure 5**.

Late Short-term (8–12 Weeks)

In the late short-term window (8–12 weeks), RI-CSI (vs Control: SMD 2.10, 95% CI 0.95 to 3.25) and SSNB+PT (vs Control: SMD=2.95, 95% CI, 2.00 to 3.91) retained the top absolute rank, while IA-CSI+PT remained the premier option among moderate-to-high certainty interventions (vs Control: SMD=1.44, 95% CI 0.47 to 2.13 in functional recovery).

Mid-term (13–26 Weeks)

At the mid-term horizon (13–26 weeks), network estimates shifted, leaving intra-articular injection of platelet-rich plasma (IA-PRP) as the absolute highest-ranked choice, representing the sole therapeutic modality with a trend of significant functional advantage (vs Control: SMD=1.16, 95% CI 0.00 to 2.33).

Late-term (27-52 weeks)

At the late-term horizon (27-52 weeks), no intervention was significant.

Table 2. Global Network Meta-Analysis Summary Across Outcomes and Timepoints

Outcome	Timepoint	Studies	Patients	Interventions	τ^2	I^2	Q Within (p)	Q Between (p)	Top-ranked (P-score)
Pain	Early Short-term (2–6 weeks)	62	4,045	16	0.565	80.3%	140.83 (<0.001)	184.87 (<0.001)	RI-CSI (0.973)
	Late Short-term (8–12 weeks)	55	3,996	16	0.776	87.7%	330.39 (<0.001)	107.76 (<0.001)	RI-CSI (0.967)
	Mid-term (13–26 weeks)	20	1,303	10	0.995	84.5%	104.95 (<0.001)	17.65 (0.001)	IA-CSI + PT (0.739)
Function	Early Short-term (2–6 weeks)	59	4,022	15	0.516	86.8%	281.05 (<0.001)	195.55 (<0.001)	SSNB + PT (0.970)
	Late Short-term (8–12 weeks)	55	4,142	16	0.475	87.0%	240.88 (<0.001)	190.86 (<0.001)	SSNB + PT (0.941)
	Mid-term (13–26 weeks)	21	1,442	11	0.523	85.0%	105.99 (<0.001)	21.03 (0.001)	IA-HD + PT (0.848)
	Late-term (27-52 weeks)	4	286	4	0.186	72.1%	3.19 (0.0739)	7.55 (0.0230)	IA-CSI + PT (0.817)

Network Meta-Regression and Subgroup Analyses

To address the global heterogeneity observed across outcomes (**Table 2**), a multivariate, 3-level random-effects network meta-regression was deployed to partition variance based on clinical and methodological diversity. Potential sources of heterogeneity were systematically analyzed by evaluating 3 major network moderators: continuous baseline pain severity, dichotomous baseline symptom duration, and procedural needle guidance methods. Model variance was partitioned into Level 3 (between-study variance) and Level 2 (within-study/between-comparison variance) to isolate specific variance drivers (**Supplemental Table S2**).

Impact of Baseline Symptom Duration and Pain Severity

Baseline symptom duration did not significantly moderate pain relief or functional recovery at early short-term or late short-term time horizons (all $p \geq 0.33$), nor did it moderate external rotation improvement at any time point (all $p \geq 0.14$). However, symptom duration exerted a statistically significant moderating effect on mid-term pain relief (Estimate = 1.47, 95% CI [0.35, 2.58], $p = 0.010$) and mid-term functional recovery (Estimate = 1.15, 95% CI [0.15, 2.15], $p = 0.025$). Contrary to the expectation that prolonged disease duration would impair treatment response, the positive coefficients indicate that patients with longer symptom duration (≥ 6 months) demonstrated a greater magnitude of improvement in both pain and function at the mid-term follow-up compared to those treated earlier (< 6 months). This pattern may reflect the natural trajectory of frozen shoulder, wherein patients with more chronic disease retain greater potential for clinically meaningful gains during the plateau or thawing phase, or may

indicate that interventions require a longer time to exert their full therapeutic effect in this subgroup.

Baseline pain severity significantly moderated early short-term pain improvement ($\beta = 0.18$, 95% CI [0.03, 0.34], $p = 0.020$), indicating that each one-unit increase in baseline pain score (0–10 VAS) was associated with a greater treatment effect at the early short-term timepoint. This suggests that patients presenting with more severe pain at baseline may derive greater early analgesic benefit from intervention. However, this moderating effect was not sustained at mid-term ($\beta = 0.02$, $p = 0.898$) or late short-term ($\beta = -0.03$, $p = 0.775$) follow-up. Baseline pain severity did not significantly moderate functional recovery or external rotation improvement across any time horizon (p values ranging from 0.42 to 0.90). Residual heterogeneity tests (QE) remained highly significant ($p < 0.001$) across all outcomes despite moderator inclusion, indicating that considerable unmeasured variance persists and that baseline pain severity alone is insufficient to fully explain the observed between-study heterogeneity.

Subgroups by Baseline Symptom Duration

In early short-term (2-6 weeks), in the chronic subgroup (> 6 months of symptom), PT-augmented combination therapies dominated pain relief. In contrast, in the subgroup with shorter symptom duration (< 6 months), RI-CSI emerged as the single dominant modality across all three outcomes, with a substantially narrower set of treatments reaching significance compared to the chronic subgroup. In the late short-term (8-12 weeks), combination therapies maintain supremacy for pain and functional improvement in the chronic subgroup; RI-CSI led the subgroup of shorter symptom duration (< 6 months). In midterm (13-26 weeks), available data was not adequate to derive conclusion. When stratified by disease duration, heterogeneity persisted across all subgroups. Taken together, the residual heterogeneity and inconsistency across subgroups indicate that disease duration alone is insufficient to explain the variability in treatment effects, and subgroup findings should be interpreted with caution.

Meta-Regression and Subgroups by Needle Guidance Method

In the subgroup of US guidance ($k = 30\text{--}32$, $I^2 = 81\text{--}95\%$), despite profound network inconsistency, RI-CSI and multi-modal PT-combined regimens (SSNB + PT, IA-CSI + PT, IA-CSI + SA-CSI + PT) dominated early short-term (2–6 weeks) across all outcomes; benefits narrowed sharply by late short-term (8–12 weeks), with only RI-CSI and SSNB + PT retaining significant functional gains.

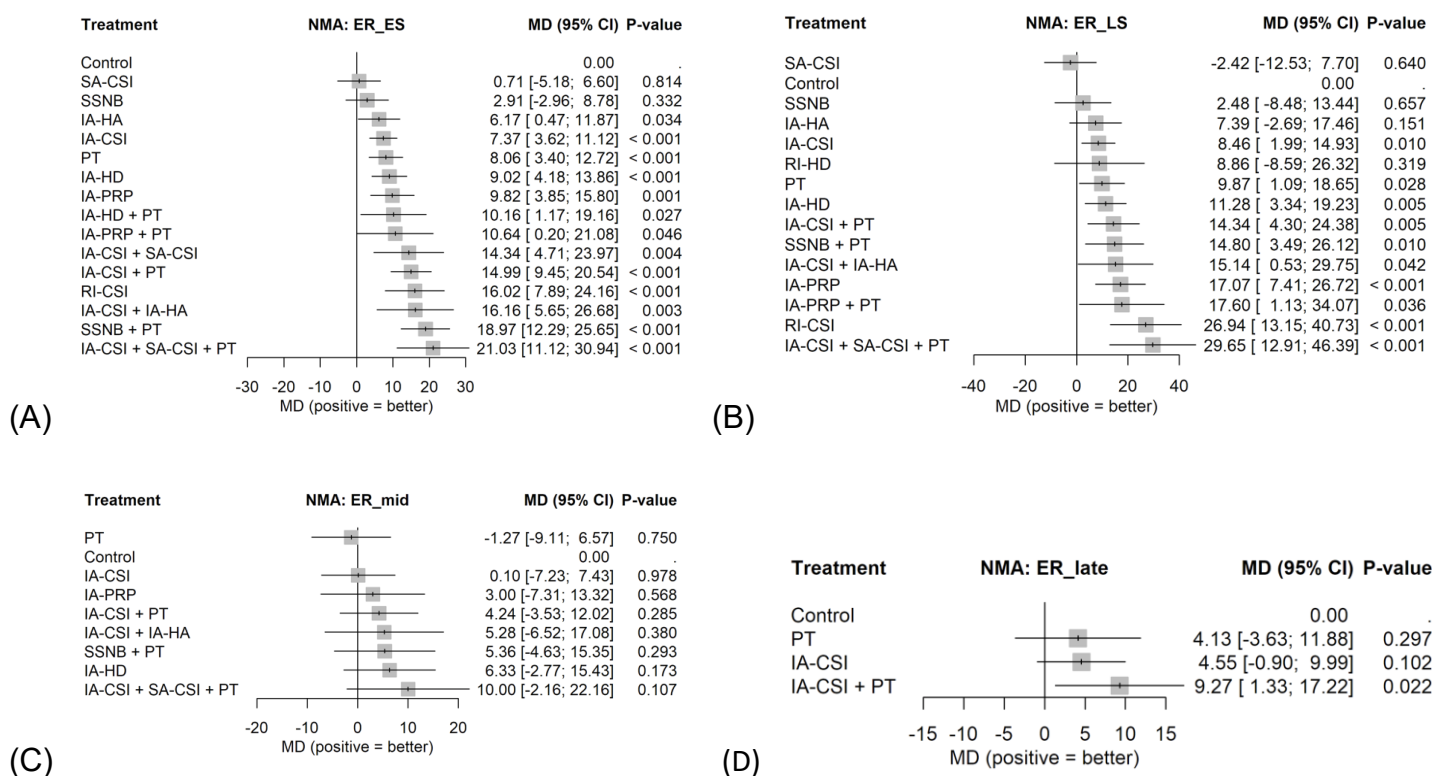
In the subgroup with fluoroscopic guidance ($k = 7\text{--}9$, $I^2 = 0\text{--}46\%$), the most internally consistent subgroup; IA-CSI + PT led pain, function, and external rotation at early short-term (2–6 weeks) and remained the only treatment with significant external rotation gains at late short-term (8–12 weeks); uniquely, IA-CSI, IA-CSI + PT, and IA-HA sustained significant pain and functional benefits into mid-term (13–26 weeks) with zero heterogeneity.

In the subgroup of landmark (blind) guidance ($k = 13\text{--}19$, $I^2 = 22\text{--}96\%$), SSNB and IA-HD + PT led early short-term (2–6 weeks) external rotation and pain, respectively; IA-CSI + PT was the only treatment consistently significant for function across early and late short-term (8–12 weeks), while IA-PRP showed durable late short-term gains in both function and external rotation; no treatment reached significance for pain at mid-term (13–26 weeks), with extreme heterogeneity ($I^2 = 96\%$) precluding reliable late short-term pain conclusions.

In the meta-regression, guidance method did not significantly moderate treatment effects across any outcome or timepoint (moderator test QM $p = 0.17$ – 0.74 for all 9 models), indicating that US, fluoroscopic, and landmark guidance do not produce systematically different overall treatment effect sizes when compared within a network meta-regression framework. US guidance showed a consistent but non-significant negative trend relative to landmark (reference) across all outcomes (all estimates negative), with the closest approach to significance observed for pain at late short-term (8–12 weeks; $p = 0.076$) and mid-term (13–26 weeks; $p = 0.085$). Substantial residual heterogeneity persisted in all models (QE $p < 0.0001$ throughout), confirming that guidance method alone explains little of the between-study variance; other unmeasured factors likely account for the majority of heterogeneity observed across guidance subgroups.

Secondary outcomes

Figure 6. Forest plot of network meta-analysis estimates for shoulder external rotation



Forest plot displaying the random-effects network meta-analysis estimates for all eligible interventions compared to the reference group (Control). Effect sizes are presented as mean differences (MD) or standardised mean differences (SMD) with 95% confidence intervals. A positive value indicates a beneficial effect (improvement) relative to Control. Treatments are sorted by effect size.

External Rotation Range of Motion

As shown in **Figure 6**, In the early short-term window (2–6 weeks), IA-CSI+SA-CSI+PT provided the most substantial absolute angular expansion (vs Control: MD=21.03, 95% CI 11.12 to 30.94). Among nodes with moderate-to-high evidence certainty, IA-CSI+PT demonstrated the most optimal and stable effect.

By the late short-term interval (8–12 weeks), IA-CSI+SA-CSI+PT still provided the most benefit for angular gains (vs Control: MD=29.65, 95% CI 12.91 to 46.39). Conversely, we found no intervention with a moderate-to-high level of certainty.

Notably, at the 13–26 weeks mid-term window, the therapeutic benefits of all active interventions waned, with no modality retaining statistical significance against control for external rotation regardless of evidence certainty.

However, during the 27-52 weeks follow-up, the therapeutic effect was significant in IA-CSI+PT, but the certainty was low due to much less studies included.

Safety

Safety outcomes, including the incidence of minor adverse events (e.g., transient post-injection pain, facial flushing, vasovagal reactions) and major complications, were compiled from reporting trials. No statistically significant differences in serious adverse event rates were observed between the active intervention nodes and conservative controls. The comprehensive safety profile matrix and specific event frequencies per interventional arm are detailed in

Supplemental Table S3.

Publication bias and local inconsistencies

A localized node-splitting analysis was executed across all network coordinates to test direct face-to-face trial metrics against indirect pathways (**Supplemental Table S4**). Significant local inconsistency ($P < 0.05$) was tightly concentrated within the early and late short-term windows across all three clinical domains. For pain relief, inconsistency was identified in the early window for the HD versus PT ($P = 0.019$) and RI-CSI versus SAB-CSI ($P = 0.021$) pathways, extending into the late window along the IA-CSI+PT versus PT axis ($P = 0.044$). For functional recovery, severe localized evidence conflicts occurred along the IA-CSI+PT versus SSNB+PT ($P < 0.001$) and the PT versus SSNB+PT ($P < 0.001$) axes. For external

rotation, discrepancies were driven predominantly by the RI-CSI versus SAB-CSI loop in both early ($P = 0.029$) and late ($P < 0.001$) short-terms.

Conversely, across all outcomes at the beyond 13 weeks of follow-up (mid-term and late-term) horizons, all localized split-tests generated non-significant boundaries ($P > 0.05$), signaling apparent network stabilization. However, because the number of contributing randomized trials dropped precipitously, the absence of identified loop inconsistency reflects a critical reduction in statistical power to detect discordance rather than a true clinical or empirical consensus between direct data and indirect pathways.

Discussion

Most Effective Injection Options

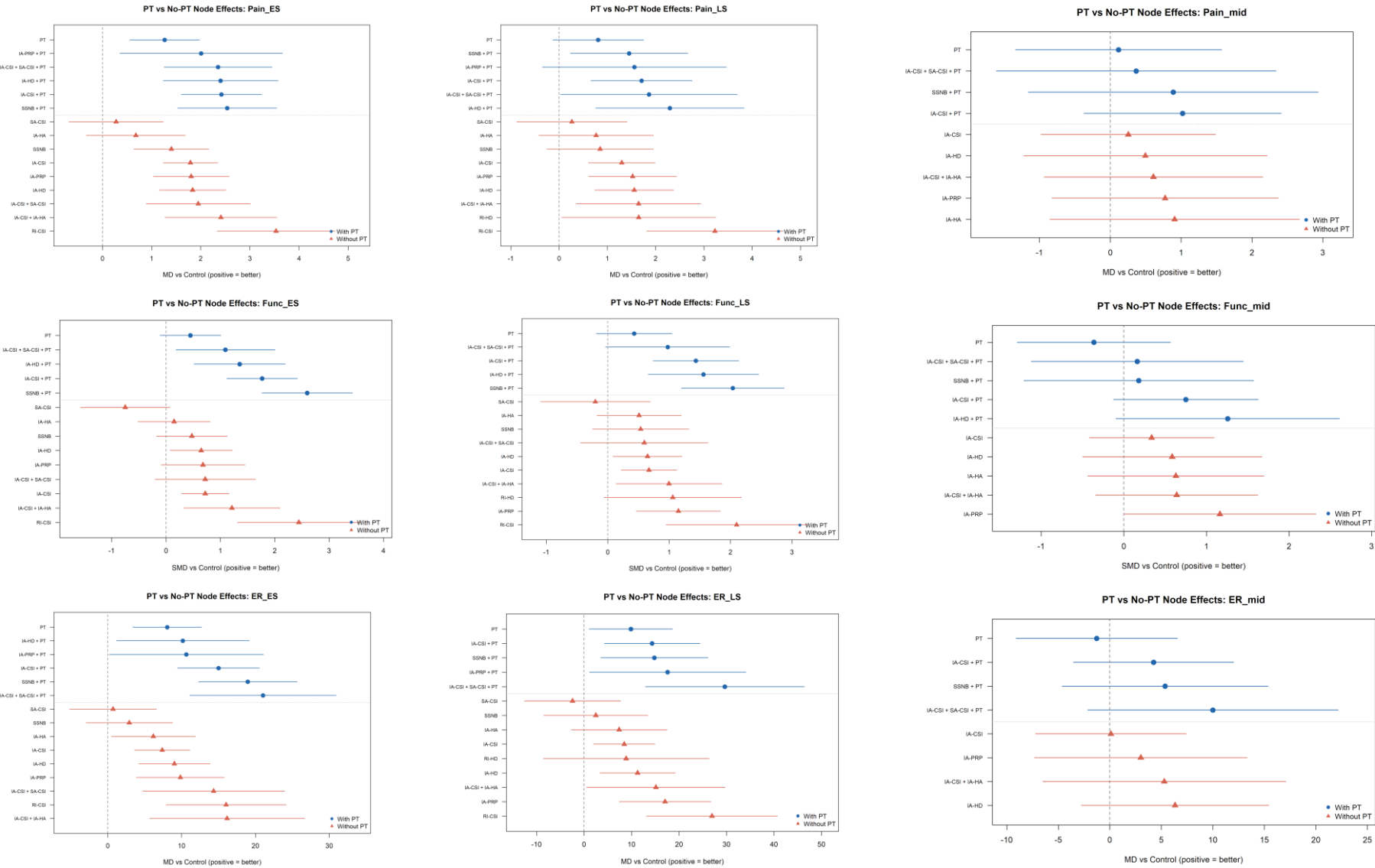
In the current study we summarized the up-to-date evidence in injection therapies for adhesive capsulitis. Injection of corticosteroid from the rotator interval was the superior treatment option for pain relief in the short-term, while in the mid-term, combination therapy (intra-articular combined with subacromial bursa steroid injection followed by supervised physical therapy) was more effective.

Intra-articular corticosteroid remains the most extensively studied and effective injection therapy for adhesive capsulitis, with consistent evidence demonstrating superiority over placebo for pain relief and functional improvement [17,18]. On the other hand, a previous study has demonstrated that capsular distension (hydrodilatation) shows strong effects, with standardized mean differences of 2.68 for pain relief and 2.89 for functional improvement compared to placebo [19]. Arthrographic distension combined with intra-articular corticosteroid demonstrates further higher probability of being the most effective treatment at 8-12 weeks than intra-articular corticosteroid alone in a previous network meta-analysis [18]. In a longer follow up duration, platelet-rich plasma (PRP) injections show benefit at 12 weeks compared with physical therapy for both pain and function, and uniquely

demonstrate benefit for shoulder range of movement [17]. However, the study number is smaller (3 RCTs, 177 participants). Like previous evidence, our study found out that in the short-term, intra-articular corticosteroid injection, hydrodilatation from the posterior glenohumeral joint capsule, and intra-articular injection of PRP are the superior treatment options which had over 5 supporting studies and clinically meaningful.

Regarding combination with nerve blocks, evidence is mixed. While suprascapular nerve blocks alone show superiority over intra-articular corticosteroids for pain at 3-4, 6-7, and 12 weeks, adding suprascapular nerve block to intra-articular steroid injection provides only transient benefit at week 2 but no clinically meaningful advantage beyond that timepoint [20,21]. Our study, on the other hand, demonstrated that in the short-term, suprascapular nerve block in combination with physical therapy provides more functional improvement than other treatments, which may be attributed to its pain-relieving effect.

Figure 7. Subgroup Based on Combination with Physical Therapy



Effects of Physiotherapy as a Positive Adjunct

We found that there is a trend of additive effect from PT especially helpful in external rotation and functional improvement during the early and late short terms. It also generated a positive effect on pain relief during the early short-term follow-up. The effects are shown in **Figure 7**.

Effects of Injection Site and Anatomy

The anatomical target of injections significantly influences outcomes. Rotator interval injection emerges as the optimal single-site approach with moderate to high certainty evidence, showing standardized mean differences of -1.07 for pain and 0.94 for global function versus placebo in short-term outcomes [22]. This approach targets the pathologically thickened capsule and rotator interval structures most commonly affected in adhesive capsulitis [23,24]. Align with previous evidence, our result suggests that steroid injection via the rotator interval provided the best pain relief outcomes spanning 2 to 12 weeks of follow-up, while the small study number warrants careful interpretation.

Multi-site injection strategies demonstrate superior efficacy. Four-site injection protocols show the most robust short-term effects (SMD=-2.20 for pain, SMD=2.02

for global function versus placebo), though with lower certainty evidence [22]. Our study also demonstrated multi-target intra-articular and subacromial bursa injection of corticosteroid in combination with physical therapy (IA-CSI + SAB-CSI + PT) provides better shoulder mobility and pain relief around 13 to 26 weeks of follow up. It is worth mentioning that the effectiveness is closely followed by corticosteroid injection via the rotator interval (RI-CSI), suggesting its proposed effect on treating multiple pathologies at one entry [23,24]. Previous studies investigating the combined effect of intra-articular and rotator interval injection showed significant benefit versus sham (mean difference -2.1 in shoulder pain at 6 weeks), though differences compared to intra-articular injection alone were not significant [25]. Further clinical studies exploring the benefits of targeting both rotator interval and posterior capsule is worth investigating.

Distension via rotator interval (RI-HD) represents the optimal approach for mid-term outcomes with moderate to high certainty (SMD=-1.10 for pain, SMD=1.46 for global function versus placebo) [22]. Importantly, distension and intra-articular injection via anterior or posterior approaches produce effects equivalent to rotator interval injection and RI-HD [22]. Our study aligned with the previous evidence, showing better effect of RI-HD (MD = 1.56, [0.08, 3.05]) than via posterior approach

in the late short-term (8-12 weeks) follow-up.

Recently evidence is growing in terms of injection into the coracohumeral ligament (CHL) and inferior glenohumeral capsule (IGHC) produces significantly greater improvements compared to posterior glenohumeral recess injection at both 2 and 4 months [24]. This aligns with pathological findings showing inflammation and edema most commonly in the CHL and anterior/inferior capsules. A previous prospective randomized trial of 120 patients showed that at 4 months, CHL+IGHC injection achieved VAS scores of 1.4 ± 0.6 versus 2.1 ± 1.1 for posterior approach, with superior range of motion improvements (abduction 176.2° vs 167.4° , external rotation 81.3° vs 72.7°) [24]. However, due to less extensive study numbers, our study did not include this intervention combination.

Effects of Corticosteroid Dose

Lower corticosteroid doses appear equally effective with potentially better safety profiles. A randomized controlled trial comparing 40 mg versus 10 mg triamcinolone acetonide for ultrasound-guided hydrodilatation found no significant differences in SPADI scores, VAS pain scores, or range of motion at 12-week follow-up [26]. Both groups experienced similar improvements throughout the study period with no adverse events reported in either group. Another meta-analysis suggests that lower dosages (10 mg vs 40 mg) of corticosteroid may lead to insignificant better pain relief and functional improvement [27]. Given the potential effects of corticosteroids on adjacent cartilage and tendons, a low dose of triamcinolone acetonide (10 mg) is preferable for ultrasound-guided hydrodilatation [26]. Based on previous analysis, we did not divide hydrodilatation with different doses of corticosteroid into separate treatment but rather prioritize the volume effect of this treatment.

Effects of Guidance

The benefit of ultrasound guidance remains uncertain. Moderate-certainty evidence indicates that ultrasound-guided injection probably provides little or no clinically important benefits compared with injection without guidance for pain (mean difference 0.5 points on 0-10 scale) or function (mean difference 2.4 points on 0-100 scale) at 3-6 weeks follow-up [28]. However, this evidence primarily derives from trials in rotator cuff disease (14 trials) with only 4 trials in adhesive capsulitis. The small absolute differences may not represent clinically meaningful improvements, and ultrasound guidance does not appear to reduce adverse event risk (RR 0.72, 95% CI 0.4-1.28). The test for subgroup differences also indicated there were unlikely to be differences across different shoulder conditions [28]. We included 25 studies of ultrasound guidance in our subgroup analysis and found similar results. While guidance methods did not generate significant difference in effectiveness, in the subgroup analysis of fluoroscopy guidance, the inconsistency was low ($I^2 = 0\%$, Q (total): 1.275, $df = 4$, $p = 0.8657$; Q (within): 1.112, $p = 0.7742$; Q (between): 0.163, $p = 0.6865$), which may be attributed to the standardization of contrast enhanced arthrography.

Effects of Pain Duration and Baseline Pain Severity Before Injection

Early intervention with corticosteroid injections is associated with shortened symptom duration and improved functional scores. In a previous meta-analysis, the subgroup analyses by disease stage reveal that steroid injection combined with physical therapy provides more benefits during the freezing phase, whereas joint manipulation provides more benefits in the adhesive phase [19]. The timing of intervention appears to be a significant moderator of treatment effectiveness, with individualized optimal intervention recommended based on disease stage [19].

Our meta-regression of the timing of intervention did not demonstrate significant differences in effectiveness except in the shoulder mobility during late short-term (8-12 weeks). Patients with prolonged symptoms (≥ 6 months) achieve a significantly lower magnitude of angular improvement than those treated early (< 6 months). Furthermore, our subgroup analysis showed that standard monotherapies dissolved into non-significance by 8–12 weeks; only mechanical capsular distension (IA-HD) remained a significant monotherapy in chronic phase of the disease (≥ 6 months of duration). On the other hand, baseline symptom duration did not significantly moderate functional recovery or pain relief across any time horizon ($P \geq 0.11$).

Effects of Baseline Pain Severity

While baseline pain severity has been identified as a potential moderating factor, the specific relationship between baseline pain levels and injection outcomes has not been systematically characterized in the available evidence. Network meta-analyses have identified adjunctive therapies, female sex, and diabetes as moderators of effectiveness, but baseline pain severity was not specifically analyzed as an independent predictor. Whether patients with higher baseline pain experience proportionally greater or lesser benefit remains unclear from current evidence [19]. Our evidence concluded that baseline pain severity demonstrated a significant moderating effect on pain improvement. However, it did not demonstrate a similar effect on ER or functional recovery.

Clinical Implications

For early-stage adhesive capsulitis, intra-articular corticosteroid injection—particularly via rotator interval approach or combined with distension—represents the most effective pharmacological intervention [17,22]. Targeting the coracohumeral ligament and inferior capsule may also provide superior outcomes compared to posterior approaches [24]. The added benefit of ultrasound guidance remains uncertain, though it may facilitate accurate anatomical targeting for specific injection sites [28]. In chronic phase of the disease, mechanical expansion of the joint capsule may be an effective choice of treatment.

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